UnionPay Integrated Circuit Card Specifications
— Product Specifications

Part VII Financial Common Application Data Store Specification

Version 2014

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1 Application Scope

This book applies to all UPI participants.

2 References

UnionPay Integrated Circuit Card Specifications – Basic Specification

ISO/IEC 7816-5:1994/Amd 1:1996 *Identification cards -- Integrated circuit(s) cards with contacts -- Part 5: Numbering system and registration procedure for application identifiers*

3 Terms and Definitions

3.1 Integrated Circuit(IC)

Electronic components used to complete the processing and/or storage function.

3.2 Integrated Circuit(s) Card

Type ID-1 cards embedded with one or more integrated circuits (as described in ISO7810, ISO7811 parts 1-5, ISO7812, and ISO7813).

3.3 Native Card

Refers to proprietary platform CPU cards, while open platforms are primarily JAVA-based.

3.4 Terminal

The device installed at the point of transaction for connection with IC card to complete financial transactions or multi-application store read-write process. This includes the interface device and may also include other components and interfaces such as host communications interface.

3.5 Application

Application protocol and relevant datasets between the card and the terminal.

3.6 Command

A message sent by the terminal to the IC card that initiates an action and solicits a response from the IC card.

3.7 Response

A message returned by the IC card to the terminal after the processing of a command message received by the IC card.

3.8 Function

A process accomplished by one or more commands, with resultant actions that are used to perform all or part of a transaction.

3.9 Message

A string of bytes sent by the terminal to the card or vice versa, excluding transmission-control characters.

3.10 Message Authentication Code

Code generated after operation of transaction data and relevant parameters. Mainly used for verification of message integrity.

3.11 Cryptogram

Incomprehensible words or signals generated by the encryption system.

3.12 Plaintext

Unencrypted data.

3.13 Key

Symbol sequence controlling encryption conversion operation.

3.14 Cryptographic Algorithm

An algorithm that transforms data in order to hide or reveal its information content.

3.15 Data Integrity

The property that data has not been altered or destroyed in an unauthorized manner.

3.16 Card Control Key (CCK)

Primary key for card, which has the highest administrative rights for the card.

3.17 Common Data Store Space

Independent storage space on financial IC cards used for consolidated storage of industry application data.

4 Document Conventions

4.1 Explanation of Terms:

When Tag appears in commands and responses, use the following symbols: M: Mandatory Tag - must appear C: Conditional Tag - appears under certain conditions O: Optional Tag - may or may not appear

4.2 Numerical Value and Format Representation

——n(numeric);

——cn(compressed numeric);

——b(binary);

——an(alphanumeric);

——ans (special alphanumeric) The following numeric symbols are used in this Specification: Decimal values are represented by the numbers themselves (ex: number 1)

Hexadecimal values are represented by adding quotation mark to the numbers, and adding x in the beginning (ex: number x'01')

Binary values are represented by adding quotation mark to the numbers, and adding b in the beginning (ex: number b'00000001' or b'1') with Bit 1 indicating the low position (0x01), Bit 8 indicating the high position (0x80)

5 Symbols and Acronyms

ADF	Application Definition File
AID	Application Identifier
APDU	Application Protocol Data Unit
CLA	Class byte of the command message
DDF	Directory Definition File
EF	Elementary File
FCI	File Control Information
INS	Instruction byte of command message
ISO	International Organization for Standardization
Lc	Exact Length of Command data sent
Le	Maximum Length of data Expected
MAC	Message Authentication Code
MF	Master File
P1	Parameter 1
P2	Parameter 2
PIN	Personal Identification Number
POS	Point of Service
PSAM	Purchase Secure Access Module
RFU	Reserved for Future Use
SFI	Short File Identifier
SW1	Status Word One
SW2	Status Word Two

6 Common Data Store Overview

6.1 Basic Overview

6.1.1 Common Data Store Overall Structure

6.1.1.1 Common Data Store Concept

Common data store refers to the pre-allocated space on IC cards, realizing the adding, reading, or deletion of applications or records through standardized data files, standardized APDU command, standardized application templates, and standardized security rules.

6.1.1.2 Common data store Applications

Common data store Application refers to using the IC card characteristic of multi-application support, pre-allocating space in the form of directory files on the IC card, and initialize according to the file interface or application template requirements in this Specification, so that this space can support subsequent application operations required by standardized commands.

6.1.1.3 Relationship Between Common Data Store Applications and UnionPay IC Applications

Common data store is an independent application; it co-exists on the same IC card with standard UICS Debit/Credit financial applications. Common data store applications and UICS financial applications are mutually independent and non-conflicting, utilizing their own self-defined internal application processes and security mechanisms. Payment related functionality on the card are carried out using financial IC card debit/credit applications. Storage format for the common data store application storage region and financial application region on IC cards are illustrated in the figure below

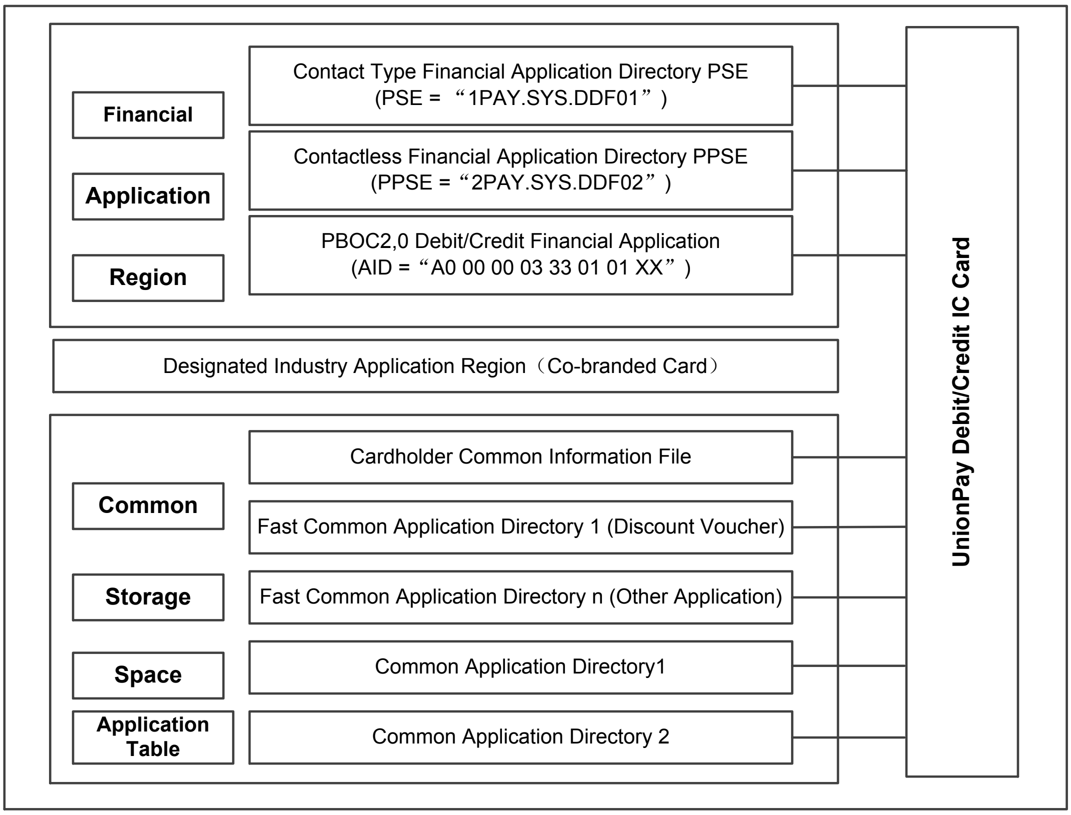


Figure 6.1. Storage format of common data store space and financial application region on IC card

——Cardholder common information file: In the common data store directory, two elementary files are saved for storing cardholder common information. These two elementary files are: cardholder common information file (EF01) and (EF02).

EF01 is a file for storing sensitive cardholder information, read after offline PIN authentication is passed, and modified with permission of the common data store space maintenance key. EF02 is a file for storing general cardholder information, with open read rights, and with edit rights under the control of the common data store space maintenance key.

——Fast common application directory: Fast common application directory can provide storage space and usage methods for some industry applications with simple data structure and low security requirements. It pre-allocates specified application directory within the common data store space. Within the application directory, corresponding industries can reference standard industry application template documents to design industry application files or records, and read or write with no verification or simple verification (for example, cardholder general storage password verification or record key control verification).

——Common application directory: Common application directory is a directory that uses more secure mechanisms for read and write control, with designated industries generally responsible for data content control and administration.

File data format and logic structures in the common application are administered by the industry itself and are not influenced by other applications. But file type and communication commands used must meet the requirements of this Specification.

——Application table: Application table is a file that mainly includes application data already added within the card and protected by card COS.

6.1.1.4 Common Data Store File and Common Data Store Industry Application Template

Common data store file is based on low-level file format defined by ISO 7816. All the various applications used in the common data store space are based on common data store files. On the basis of common data store files, this Specification also defines some standard industry application templates (can be application or file) based on various industry characteristics, so relevant industries can use these templates to write into common data store region.

6.1.2 Security Mechanism

IC cards that meet this Specification must satisfy the following basic security mechanisms: Application security: Includes application and inter-application security, security added or removed by application, etc. Data security: Security control over application data read, write, and edit. Authentication security: Card and terminal bi-directional authentication, terminal authentication of card, card authentication of system backend, system backend authentication of card, etc.

6.1.3 Re-Personalization

Cards that meet this Specification must support modification of common data store space content after issuance by the issuer, using specified standardized commands, through adding an application or a common data store file. Generally the basic flow is as follows:

Step 1: Select common data store space Step 2: Read card random number Step 3: Use authentication command for authentication Step 4: write common application or common data store data

6.1.4 IC Card Physical Requirements For Supporting Common Data Store Functionality

In order to realize multi-application storage on the card, cards that meet this Specification is recommended to possess at least the following characteristics:

——Central processing unit: 8bits high performance CPU (or higher)

——Co-processor: DES/3DES, random number generator

——Data storage space: 32K EEPROM/FLASH (or higher)

——Program storage space: 64K ROM/FLASH (or higher)

——Data retention time: 10+ years

——Data region overwrites: 100K+ times

——IO communication frequency: contact interface 9600+ bps, contactless interface 106+ Kbps, and meets UICS specification requirements.

7 Common Data Store Space and Common File Definition

7.1 Common Data Store Space

7.1.1 Common Data Store Space Creation

Issuer creates common DDF in the process of issuing cards, all common data store applications based on the Specification should be stored under this DDF, and can be operated through standardized APDU command, the space for this DDF on the IC card is set upon issuance and could not be changed. Method of creating DDF can be self-defined, for example, JavaCard creates DDF using install method.

Common data store space and other card space are mutually independent physically and logically. Common data store space operations do not impact any applications or data in other spaces. To ensure secure separation between different applications on the same card, each application should be placed in an independent ADF, with logical firewalls set up between applications to prevent illegal cross-application access, while also ensuring that each application has no conflicts with other application rules stored on the card. This Specification does not restrict relative order in card application creation, recommend following a principle of consecutive physical space allocation.

7.1.2 Initialization of Common Data Store Space

7.1.2.1 Space Initialization

Common data store space initialization includes basic space layout (AID, names, etc.), read-write rights configuration, user offline access password initialization setup, etc.

7.1.2.2 Space Basic Information

Common data store space has the following characteristics at creation time, with 0s added at the end if there are not enough AID digits.

Table 7.1.   Common data store AID

AID	DDF Name	Size	Other
A000000333CDD000	Common data store	Self-defined	Reserved

7.1.2.3 Read-write Rights Configuration

Common data store space are created under the control of the card control key, with relevant read and write application rights configured after creation, while also simultaneously generating the application maintenance key for the space to maintain subsequent applications.

Common data store space also should establish relevant offline password authentication rights, with this offline authentication password used mainly for read-write management of some fast common industry application directories and cardholder common information files.

7.1.3 Common Data Store Space Structure

Common data store space generally should include 2 cardholder general information file, 1 common application table file, 1 or faster common application directory (ex. discount voucher, point accumulation), several specific common applications, and space required for adding additional applications, or other specific industry applications determined at issuance. Figure 7.1 is an illustration of common data store space file and application structure.

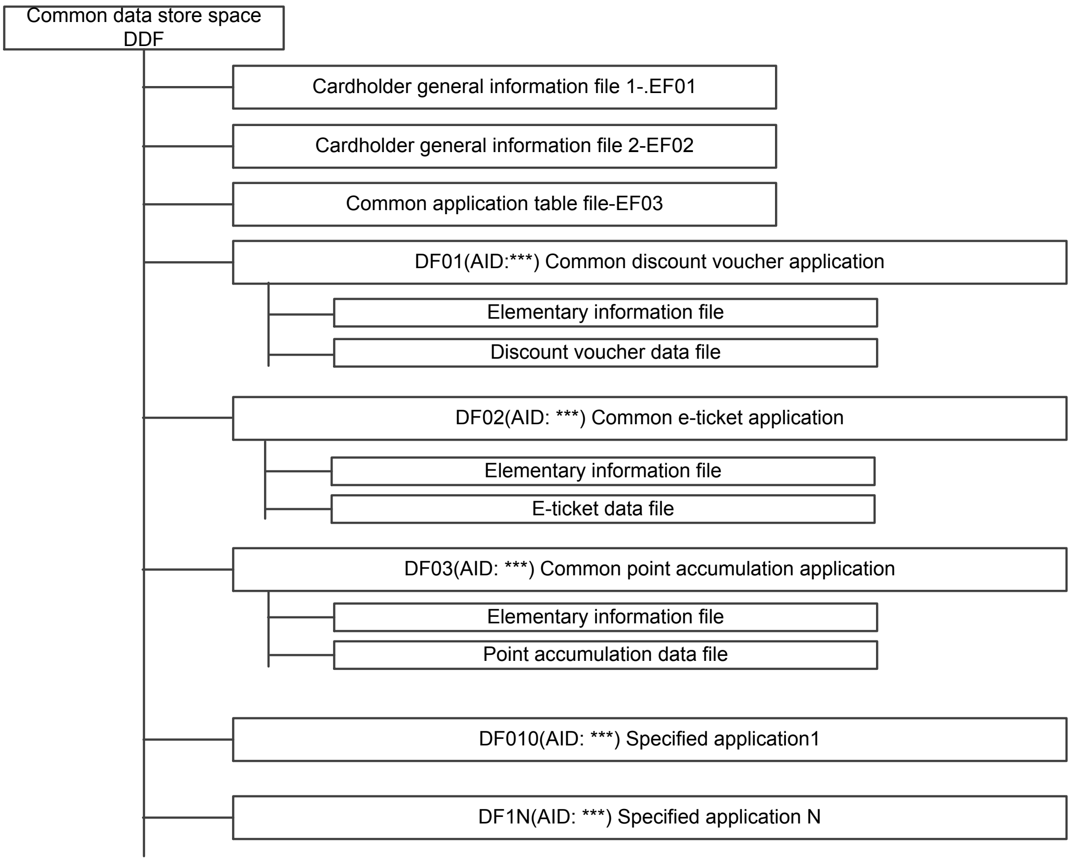


Figure 7.1. Structural map of common data store space and applications

7.1.4 Common Data Store Space Options

When the terminal chooses common data store space content, after successful execution of select command, PSE, DDF, or ADF routes on the IC card are set, subsequent commands will use the AEF related to the PSE, DDF, or ADF determined by the SFI.

Response message after card executes SELECT DDF is as follows:

Table 7.2. Response message for SELECT DDF (FCI)

Tag			Value	Mode
‘6F’			FCI template	M
	‘84’		DF name	M
	‘A5’		FCI data proprietary template	M
		‘88’	SFI for directory elementary file(EF03)	M

Table 7.3. Response message for SELECT ADF (FCI)

Tag			Value	Mode
‘6F’			FCI template	M
	‘84’		DF name	M
	‘A5’		FCI data proprietary template	M
		‘’50’	Application tag	M
		‘9F08’	Common data store version number	O
		‘BF0C’	Issuer discretionary data(FCI)	O

Tag		Value	Mode
	'xxxx'	One or more additonal (proprietary) data elements from the application supplier, issuer, or IC card supplier	O

IC card should support terminals obtaining card information through the Get Data command defined in section 8 of this Specification (ex. remaining space size, status), so as to facilitate merchant partner or specified application supplier determining relevant state when adding application. The remaining space value returned by the IC card is for application supplier reference only, it does not mean applications of that size can be completely added.

7.1.5 Common Data Store Space Authentication

Accessing common data store space doesn't require authentication in and of itself, but creating internal data files and directories require external authentication and is executed under the protection of the common data store space control key.

7.2 Common Information File Content

7.2.1 Common Information File Content Overview

Common information file mainly includes two types of content, one being the cardholder general information file. The second is the common application table information file.

7.2.2 Cardholder General Information File

Cardholder general information file includes two elementary files, which store cardholder general information. The two elementary files are: cardholder general information file (EF01) and cardholder general information file (EF02).

Cardholder General Information File (EF01):

EF01 is a file for storing sensitive cardholder information, read after offline PIN authentication is passed, and modified with permission of the common data store space maintenance key.

Table 7.4. Cardholder General Information File (EF01)

EF01: Cardholder General Information File		
ID: EF01	Type: Binary file	SFI: 01H
Read Right: PIN		
Write Right: Space maintenance key		
Data item	Offset	Length
Cardholder sensitive information	0	64

Recommend sensitive information should include but be not limited to the relevant content in the table below:

Table 7.5. Sensitive information format

No.	Data Item	Format	Length
1	Name	an	20
2	Proof of Identification Type	cn	1
3	Proof of Identification No.	an	20
4	Contact No.	an	16
5	Reserved		7

Data format filling rules are as follows:

Table 7.6. Data format filling rules

When the defined data length is greater than the actual data length, and not all positions are filled, then padd as follows: ——Format n data right align, left fill with 0; ——Format B data left align, right fill with 0; ——Format cn data left align, right fill with F; ——Format an data left align, right fill with 0; ——Format ans data left align, right fill with 0.

Cardholder general information file (EF02) stores cardholder commmon information, with open read rights and with common data store space maintenance key controlling revisions. Formats are as in the table below:

Table 7.7. Cardholder general information file EF02

EF02: Cardholder general information file		
ID: EF02	Type: Binary file	SFI: 02H
Read Right: Free		
Write Right: Space maintenance key for revision		
Data item	Offset	Length
Cardholder common information	0	72

Recommend common information to include but not be limited to content in the following table:

Table 7.8. Common information content

No.	Date Item	Format	Length
1	Virtual card no.	an	30
2	Other card no.	cn	30
3	Bank code	an	8
4	Reserved		4

The virtual card number can be used for membership number or other scenarios that require card unique id code. Virtual card number configuration rules are:

Member institution id code (8 digits) + region code (8 digits) + 14 digit serial number. For member institution number rules see "China UnionPay Member Institution ID Code Specification".

For region code see "UnionPay Card Cross-Industry Business Region Code Standard", pad with 0s as needed on the right if fewer than 8 digits.

Serial numbers are defined by each issuer, but shall not exist in duplicate.

Other card number can be set as cardholder actual card number.

7.2.3 Common Application Table Information File

Common application table information file (EF03) is a file that mainly includes application data already added within the card, with open read rights, and protected by card COS. It can provide cardholder with information of already added applications. This file preserves the basic information for all applications (including fast directories and specified directories) currently in the common data store space. When executing add and delete operations on applications in the common data store space, card COS must automatically maintain this information file in order to complete the add and delete operation on the corresponding directory information, and can obtain currently valid information through the Read Record command. Any deleted application directory records and non-valid records would not be read out.

The file ID of the common application table file is EF03. For Read Record EF03 the card follows PBOC2.0 Specification Part III (Payment System Directory Code

Definitions) to return 70 template record, with each record being the record format for an application gateway information file.

7.3 Fast Common Application Directory

7.3.1 Fast Common Application Directory Concept

For some industry applications with clear data structure and relatively low grade security requirements, common data store space can be pre-categorized and allocated specific application directories, or fast common application directories. Files in the directory could be written in according to pre-set industry common data store application templates.

7.3.2 Fast Common Application Directory Creation

The creation of fast common application directories requires having passed external authentication to enable creation in the common data store space, issuer can create this during card issuance, or can create onsite or remote through terminal or network after issuance. Creation of fast common application directories must follow the AID allocation principle standardized in this Specification.

7.3.3 Fast Common Application Directory Structure

In the fast common data store directory, one or more files (may include counters, key files, information files, etc.) based on industry characteristics. File types used must meet specification definition and can be personalized through referencing common data store industry application templates.

7.3.4 Fast Common Application Directory Initialization

When creating fast common data store directories, the size limit of the space taken up must be specified. When creating files in the fast common data store directory, the read-write protection rights of the files must also be set.

7.3.5 Fast Common Application Directory Options

Unless under the control of the common data store space control key or the control key of the current directory, the fast common data store directory could not delete, but rather only has read rights with the ability to return directory information, directory tables in files, etc. (ex. discount voucher information). After choosing a fast common data store directory, only the files within can be operated on. Files in other directories could not be operated on.

7.3.6 Adding Fast Common Application Directory

Fast common data store directories that are not created during initial card issuance could be added to the space afterwards after completing development under the control of the common data store space control key or add application key.

7.4 Common Application Directory

Mainly refers to specific industry applications that follow standardized commands, differing from fast common applications usually due to higher grade security requirements, requiring industry control to open files for records, with file structure self-defined by application provider. Its AID can be allocated centrally by UnionPay, so that the same industry would only need to develop one set of industry applications to meet all bank card applications.

Common application directory creation is similar to that of fast common application directory.

7.5 File Types Used in Common Data Store Space

7.5.1 EF File Supported by Common Data Store Space

In the common data store space region, the following EF file must be supported: binary file, variable length file, fixed length record file, circular file, counter file, and fixed length record file with extension field. EF files can only be searched for through SFI (the lower 5 bits of the last byte of the FID), with the SFI specified as the P1 or P2 parameter in the access command. The valid range of SFI is 1~30, so under any DF there could be a maximum of 30 EF files.

7.5.2 Binary File

Binary files are also known as transparent files, with file body being a consecutive region and bytes as the unit of access, address selection must be through offset and length. This offset is specified in the binary file access commands Read Binary and Update Binary, and reading or updating binary files requires meeting file read/write rights and mode control. The offset for the first byte in the binary file is 0, the offset for the second byte is 1, and so forth.

The structure of binary files is as shown in the figure below:

0123456789ABCDEF0123456789ABCDEF…...

Figure 7.2 Binary file structure

Binary files are filled with 0x00 by COS upon creation. Since binary files can only be accessed through SFI, the max file length is limited to 510 bytes. If there is a significant amount of data to store, then using record file or creating multiple binary files should be considered.

7.5.3 Variable Length Record File

Length of records in variable length record files may vary. Once a record is created or added, its length may not be changed. Variable length record EF files consist of sequential record data, from record #1 to record #N (with N being the number of records in the file), each record being a complete object. Reading or updating both use a single record as the unit, and updating a record does not change its record number.

Variable length record files can use "Append Record" commands to add records. After adding the records, it could be updated through the "Update Record" command, and read through the "Read Record" command. Reading or updating variable length record files requires meeting file read/write rights and mode controls.

The structure of variable length record files is as shown in the figure below:

Record #1
Record #2
…
Record #N

Figure 7.3 Variable length record file structure

7.5.4 Fixed Length Record File

Each record in a fixed length record file has the same length, while other characteristics are identical to that of variable length record files. The various records of a fixed length record file are pre-set by COS at creation time, filled with 0x00, updated through the "Update Record" command and read through the "Read Record" command.

Reading or updating fixed length record files requires meeting file read/write rights and mode controls.

The structure of fixed length record files is as shown in the figure below:

Record #1
Record #2
…
Record #N

Figure 7.4 Fixed length record file structure

7.5.5 Circular File

The length of each record in a circular record file is the same, supporting circular record data access, with record #1 being the latest written, and record #N being the earliest written. When the EF file is filled up, the record number of a newly added record is #1, generally referred to as "current record", while the last record is deleted. Other records have sequence number incremented by 1.

Circular record files have records pre-set by COS at creation and filled with 0x00. New records can be added through the "Append Record" command or cover the earliest written record when filled up. The last added record has record number 1, and is the newest record; designated records are updated through the "Update Record" command, and read through the "Read Record" command. Reading or updating circular record files requires meeting file read/write rights and mode controls.

The structure of circular record files is as shown in the figure below:

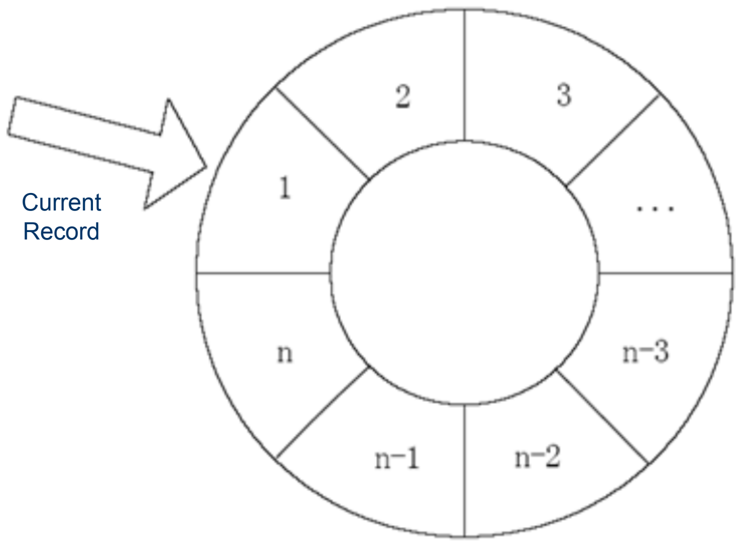


Figure 7.5 Circular record file structure

7.5.6 Counter File

Counter EF file consists of N counters. Each counter has 4 bytes of counter value (can represent unsigned number range of 00h~FFFFFFFFh, or 0~4294967295).

The value of each counter can be read through the Read Count defined in Section 8. Add, subtract, or reset operations are through Increase Counter and Decrease Counter. All operations require MAC. The structure of counter files is as shown in the figure below:

Counter #1	**Counter #2**	**Counter #3**	**…**	**Counter #N**
xx xx xx xxh	yy yy yy yyh	zz zz zz zzh		nn nn nn nnh

Figure 7.6 Counter file structure

7.5.7 Fixed Length Record File with Extension Field

Fixed length record file with extension field is a type of file extended on the basis of fixed length record files. Each record in a fixed length record file with extension field has its own maintenance key (16 byte long 3DES key). This key is used for security protection during delete and update operations on the record. The data body length of each record inside a fixed length record file with extension field is recommended to be no longer than 230 bytes.

Fixed length record files with extension fields are pre-set with records and initialized with 0x00 by COS at creation time. The records in fixed length record files can be read with the Read Record command defined in Section 8 (keys cannot be read). Adding, updating, and deleting records can be done through Append E-Record, Update E-Record, and Delete E-Record. Access requires fulfilling the corresponding security requirement.

Record #1 Data Object	Record #1 Maintenance key protection mechanism
Record #2 Data Object	Record #2 Maintenance key protection mechanism
…	…
Record #n Data Object	Record #n Maintenance key protection mechanism

Figure 7.7 Fixed length record file with extension field structure

Each record above corresponds to a maintenance key protection mechanism. This protection mechanism may store a key after the record, or may correspond to a key id with the key stored within a key file.

7.6 Common Data Store Industry Application Template

Common industry application templates are mainly meant for defining and explaining point accumulation files, discount voucher files, and other commonly used application structures. When a party in the relevant industry initiates application development, corresponding data could simply be added to the industry template, with no need for re-defining. This industry application template can also be amended and added to subsequently. For details please see the sample application in the Appendix.

7.7 Fast Common Application AID Allocation Mechanism

Application Identifier (AID) structure must meet *ISO/IEC 7816-5:1994/Amd 1:1996*, utilize BCD encoding with 16 byte 32 digit numbers, including two parts:

(I) RID (Registered Application Provider Identifier): A registered application provider id, identifies a unique application provider. Length is 5 bytes. For all China UnionPay applications this is standardized to be A000000333.

(II) PIX: Proprietary Application Identifier Extension. Its length varies between 0 to 11 bytes, for a max of 11 bytes. Self-defined by application provider, with value determined by application provider. The meaning of this field is only for specified RID, the PIX for different RID do not need to be unique.

Current fast common applications has been categorized, with defined AID and name as follows:

Table 7.9. Common application AID categorization

AID	ADF Serial No.	Application Type	ADF Name	Rights
A000000333CDD001	DF01	Discount voucher	Common discount voucher	Read free Write Offline PIN
A000000333CDD002	DF02	E-ticket	Common discount voucher	Read free Write Offline PIN
A000000333CDD003	DF03	Membership card application	Common membership card	Read free Write Offline PIN

Other specified fast common applications: serial no. DF04-DF09 AID and specified application names will be added for use after new application templates are confirmed.

This Specification current only covers fast common application directory AID. AID that are self-defined by each org are not managed centrally, so relevant organizations should avoid overlap with Specification when determining own AID.

7.8 Space Recycling after Application Deletion

Since common data store space could readily create and delete applications (and internal files), it is inevitable that some space fragmentation will result. In order to effectively and reasonably utilize common data store space, cards must have some space recycling mechanism. The recycling mechanism is in essence a process of organizing and reusing space fragments. After numerous allocation and recycling processes, the large open sectors on the card gradually become divided into small used sectors. In order to more effectively use common data store space, it is recommended that neighboring space fragments should be merged for subsequent easy of use. Specific implementation mechanism for space recycling is not covered by this Specification, but left to independent design by developer.

8 Common Data Store Security Mechanism

8.1 Common Data Store Security Control

Since common data store regions, applications and files have different data security requirements, a clear security system must be established in the common data store space, determining the corresponding security protection grade.

8.2 Common Data Store Key Category

8.2.1 Key Category

According to common data store region file structure, in order to complete relevant application operation and maintenance, IC card must have set up keys as follows.

Table 8.1 Key category

No.	Key Name	Key Type	Key Description
1	Application Control Key (DACK, ACK)	00	The highest level key under DDF or ADF, used for control administration of the whole application (including application addition and deletion), as well as the protection and updating of other keys. Control key id is fixed as 0x00, created automatically upon creating ADF or DDF, and updatable.
2	Application Adding Key (DAAK)	00	Used to protect the addition of ADF under DDF, as well as the protection and updating of other keys. This key can only exist under DDF, key id is fixed as 0xFF, automatically created upon creation of DDF, and updatable.
3	PIN Reset Key (DPRK)	01	Resets blocked PIN, only exists under DDF. Key id range is 0x00

No.	Key Name	Key Type	Key Description
4	Application Maintenance Key (DAMK, AMK)	01	Used for secure write operation in application files, to calculate either the update command MAC or encrypt the content that needs to be updated. Key id range is 0x01~0x55.
5	Counter Increment and Decrement Key (INCK, DECK)	01	Used for incrementing and decrementing counter. Key id range is 0x56~0xFF
6	External Authentication Key (DEAK, AEAK)	02	Key used for external authentication. After passing external authentication, can obtain the access rights for relevant files. Key id range is 0x01~0xFE.
7	Internal Authentication Key (DIAK, AIAK)	03	Key used for terminal authentication of card
8	Common data store user PIN	04	Obtain relevant application or file access right through PIN authentication, only exists under DDF. For example, access to fast common application can be implemented using PIN

Issuance administration and mutual protection relationship of the keys above are as in the table below:

Table 8.2 Key protection relationship

Key Name	Acronym	Creator	Administrator	Storage Location	Protection Relationship
Common data store region control key	DACK	Issuer	Issuer	Common data store region in DDF	Protected by CCK during creation, after successful write-in provides its own protection
Common data store region add application key	DAAK	Issuer	Issuer	Common data store region in DDF	Protected by DACK
Common data store region application maintenance key	DAMK	Issuer	Issuer	Common data store region in DDF	Protected by DACK
Cmmon data store region PIN reset key	DPRK	Issuer	Issuer	Common data store region in DDF	Protected by DACK
Common data store region external authentication key	DEAK	Issuer	Issuer	Common data store region in DDF	Protected by DACK
Common data store region internal authentication key	DIAK	Issuer	Issuer	Common data store region in DDF	Protected by DACK

Key Name	Acronym	Creator	Administrator	Storage Location	Protection Relationship
(fast) Common application control key	ACK	Issuer	Application Provider	Industry application in ADF	Protected by DACK or DAAK during creation, after successful write-in provides its own protection
(fast) Common application maintenance key	AMK	Application Provider	Application Provider	Industry application in ADF	Protected by ACK
(fast) Common application external authentication key	AEAK	Application Provider	Application Provider	Industry application in ADF	Protected by ACK
(fast) Common application internal authentication key	AIAK	Application Provider	Application Provider	Industry application in ADF	Protected by ACK
Increment key, decrement key	INCK DECK	Application Provider	Application Provider	Industry application in ADF	Protected by ACK

Note: The keys above are 128 bits long.

Overall security system is as shown in the figure below:

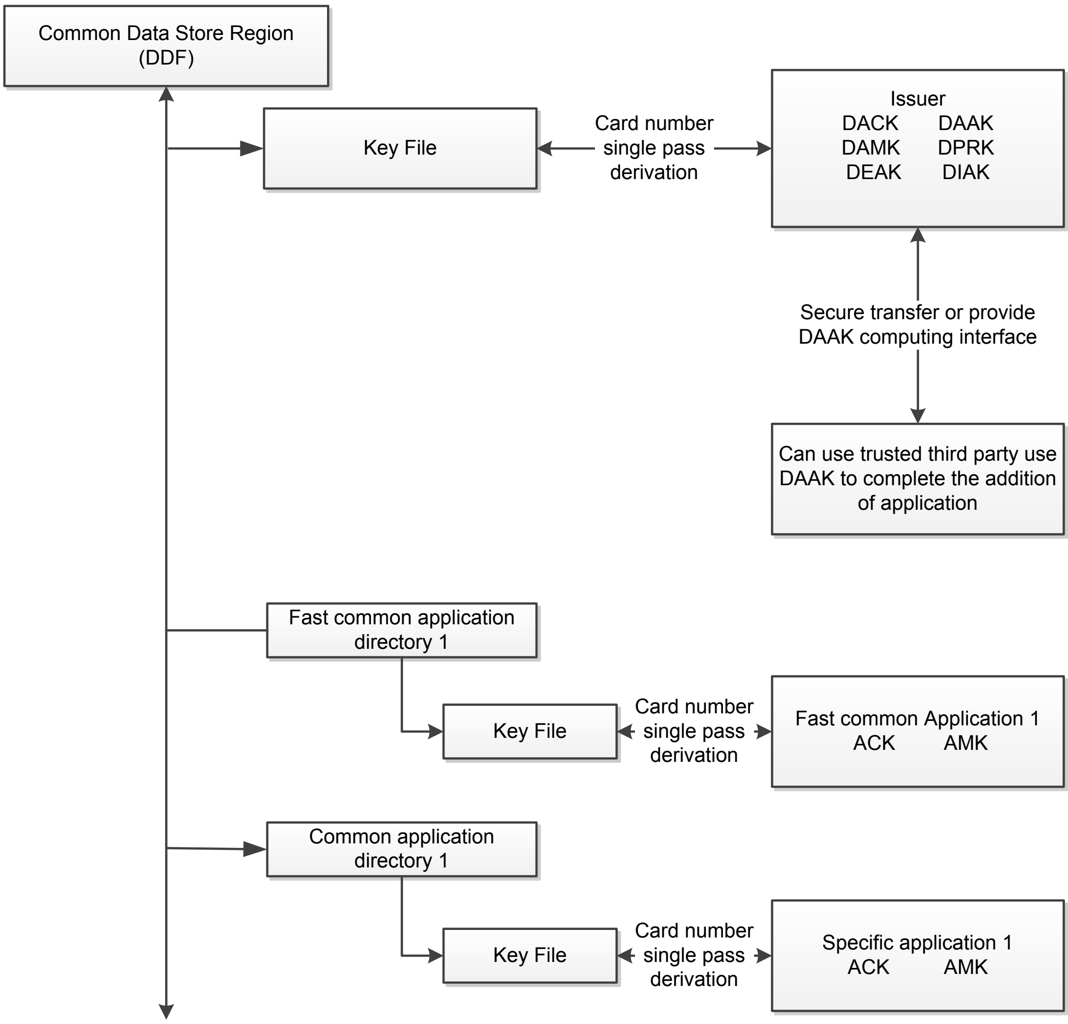


Figure 8.1 Complete key security system

8.2.2 Common data store user PIN

Common data store region PIN is controlled by the user, set upon issuance, and could be modified subsequently. This PIN is mainly used for some low-level security applications, files, and records for simple authentication. For example, passing PIN authentication enables the addition and deletion of fast common application directory records, reading of sensitive information, search for common data store region applications or fast common application directory records, and other operations.

Common data store region PIN must be securely stored in the IC card, and not be disclosed under any circumstance. Stored separately from financial PIN, PIN resets must be protected by the PIN reset key.

8.2.3 Common Data Store Maintenance Key

Common data store region maintenance key is used to maintain the common information files stored in the common data store region, generally managed by the issuer.

8.2.4 Common Data Store Region PIN Reset Key

Common data store region PIN reset key is used for authorization and protection during the process of resetting common region PIN. This is maintained by the issuer.

8.2.5 Common Application Control Key

Common application control keys are the highest level keys in the (fast) common application ADF, used for the administrative control of the entire application (including the addition and deletion of files), as well as the protected update of other keys. This control key can be established by the industry organization when writing in the common application, or can be controlled by the bank.

8.2.6 Common Application Maintenance Key

Common application maintenance key is used for the maintenance of the following files in (fast) common applications, for example controlling the read-write rights of files, MAC verification, etc. This key can be established by the industry organization when writing in the common application, or can be controlled by the bank.

8.3 Key Application Corresponding Relationship

The overall common data store region uses a hierarchical control mechanism, the corresponding relationship of relevant keys in the specific application process are as shown in the table below:

Table 8.3 Key application corresponding relationship

Application Type	Application Name	Create	Delete	Update	Read
Directory	Common Data Store Region	CCK	CCK	—	—

Application Type	Application Name	Create	Delete	Update	Read
	(Fast) Common Application Directory	DAAK or DACK	DACK	—	—
File	Cardholder General Information File	DACK	—	DAMK	PIN (sensitive data), others open
	Common Application Table Information File	—	—	—	Read open, maintained by COS
	Fast Common Application Data File	ACK	—	AMK	Read open, record files with protective key needs to pass protective key to read
	Common Application Data File	ACK	—	AMK or need to pass external authentication	According to application provider needs
	Counter File	ACK	—	INCK DECK	Read open

Application Type	Application Name	Create	Delete	Update	Read
PIN	Common data store user PIN	DACK	—	DPRK	—

8.4 File Access Control

File access rights include the read right and write right of files. They define the operating conditions of files. For binary files, fixed length record files, variable length record files, and circular record files, in order to execute read/write operations on the file, the read-write rights of the file set at file creation time must be met.

For fixed length record files with extension field, in order to execute read/write operations on records in the file, the corresponding read/write rights of the specified record must be met. After file creation, the read-write rights of the file could not be modified. File access conditions have the following three choices for security mode:

1. Open access: no restrictions on file access

2. PIN protected access: to access file, must first pass PIN authentication

3. Secure state protected access: to access file, must pass specified key external authentication. Access condition of 2 and 3 could be used in combination. File data exchange mode offers two choices:

1. Plaintext+MAC read/write: to read or write file, must pass specified key MAC verification, and use plaintext mode for read/write.

2. Cryptogram+MAC read/write: to read or write file, must pass specified key MAC verification, and use cryptograms for read/write.

8.5 Security Message

8.5.1 External Authentication Cryptogram Calculation

In common data store applications, external authentication cryptogram length is 8 bytes, and the method of calculating external authentication cryptogram is as follows: Prior to external authentication, need to pass random number command, obtain 8 byte random number; use DEAK to execute 3DES calculation on the 8 byte random number entered (ECB chain mode; no fill processing) to obtain the cryptogram data required for external authentication, with the external authentication command data sent to card for authentication.

8.5.2 MAC Calculation

Prior to each MAC calculation, random number must be obtained as initial vector. Specific calculation steps are as follows: Common data store uses DES algorithm with filling as the encryption method for calculating MAC. MAC calculation requires 5 steps:

Step 1

Take 4 bytes of random number + '00 00 00 00' as initial variable. Step 2 Follow sequential order to connect the following data into a single data block:

CLA, INS, P1, P2, Lc (Lc being the actual length of the command data field sent by the terminal, including 4 byte MAC), specified application fixed data (plaintext or cryptogram data included in the command data field)

Step 3

Break up the large data block created in step 2 into 8 byte unit small data blocks, labeled as D1, D2, D3, and D4, etc. The last data block may have 1-8 bytes.

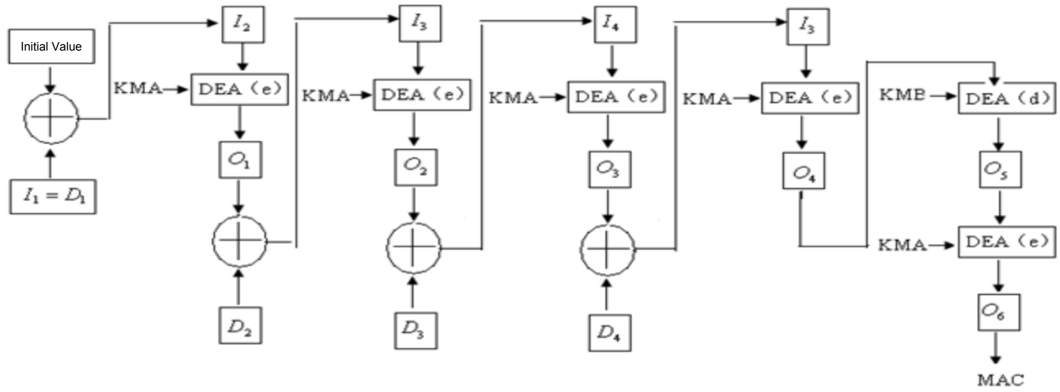
Step 4 Fill in data to the last block as follows, to make it 8 bytes: If the last data block is already 8 bytes in length, then append 8 bytes of hexadecimal '80 00 00 00 00 00 0000'

If the last data block has less than 8 bytes, then append hexadecimal '80', if still less than 8 bytes, then append hexadecimal '00', until it is 8 bytes long.

Step 5

For these 8 byte long data blocks use the designated security message key (application maintenance key or counter increment and decrement keys, etc.), use DES algorithm to encrypt (see Figure 7.2).

The number of 8 byte data blocks to be calculated using this method depends on the length of the data block calculated in step 2. SM-MAC is the first 4 bytes (from left side) of the derived 8 byte data encrypted block.



I=Input Å=XOR Operation

O=Output DEA(e)=data encryption algorithm (encryption mode)

D=data block DEA(d)=data encryption algorithm (decryption mode)

KMA=double length key left 8 bytes

KMB=double length key right 8 bytes

Figure 8.2 Double length DEA Key MAC calculation

8.5.3 Data Encryption

Reference UnionPay Integrated Circuit Card Specification – Basic Specification Part 4 for relevant content.

8.5.4 Key Derivation

Reference UnionPay Integrated Circuit Card Specification – Basic Specification Part 4 for relevant content.

9 Common Data Store Standardized APDU Command

9.1 APDU Command and Response Table

The reference field in the table below indicates the type of command, if ISO/IEC 7816-4 is indicated, then that means it is a standard command, otherwise it is a proprietary command.

Note: Commands with CLA lower half byte being '4' must have MAC in its data field.

Table 9.1 APDU command table

CLA	**INS**	**Name**	**Reference**
80/84	E0	Create File	
80	0E	Delete DF	
80	5C	Read Count	
84	5D	Increase Count	
84	5E	Decrease Count	
80	CA	Get Data	
84	D4	Write Key	
00	84	Get Challenge	ISO/IEC7816-4
00	88	Internal Authenticate	ISO/IEC7816-4, allows modification
00	82	External Authenticate	ISO/IEC7816-4
00	B0	Read Binary	ISO/IEC7816-4

CLA	INS	Name	Reference
00/04	D6	Update Binary	ISO/IEC7816-4
00/04	E2	Append Record	ISO/IEC7816-4
00	B2	Read Record	ISO/IEC7816-4
00/04	DC	Update Record	ISO/IEC7816-4
00	A4	Select File	ISO/IEC7816-4
00	C0	Get Response	ISO/IEC7816-4
00	20	Verify PIN	ISO/IEC7816-4
80	5E	Change PIN	
80	5E	Reload PIN	
84	E2	Append E-Record	
84	D8	Update E-Record	
80/84	D7	Delete E-Record	
84	1E	Application Block	
84	18	Application Unblock	

Description of main command status words are as in the table below:

Table 9.2 Command status work description

SW1	SW2	Meaning
90	00	Command executed successfully
61	XX	Command executed successfully, has XX bytes that needs to be retrieved through Get Response command
63	CX	Authentication failure, has X tries remaining
64	00	Non-volatile memory state change, command processing complete but not correct
65	81	Memory failure, non-volatile memory state change, command processing failure
67	00	Length error
69	01	Can not process
69	81	Command and file structure mismatch
69	82	Does not meet security status
69	83	Authentication method blocked
69	84	Random number not requested
69	85	Usage condition not met
69	88	Security message (MAC) error
6A	80	Data field parameter error
6A	82	File not found

SW1	SW2	Meaning
6A	83	Record or counter not found
6A	84	Insufficient space
6A	86	Parameter P1, P2 error
6B	00	Offset incorrect
6C	XX	Use parameter Le=XX to resend command
6D	00	Wrong INS
6E	00	Wrong CLA
6F	00	GET RESPONSE command has no data to use
94	02	Counter overflow, post-transaction amount less than 0 for greater than 0xFFFFFFFF
94	03	Key not found

9.2 APDU Command Details

9.2.1 Create File: Command to Create File

——Command description: This command is used to create common data store files, including ADF, EF (binary files, record files, counter files, record files with extension field, and security files used to store key data). Note: DDF creation is not defined in this Specification.

——Usage conditions and security:

1) Creating file under the common data store space requires authentication through the application control key (DACK) or the add application key (DAAK) of the DDF; if command requires MAC then corresponding key calculations should also be used;

2) Creating file under the ADF requires authentication through the application control key (ACK) of the ADF, if command requires MAC then ACK key calculations should also be used;

3) ACK must be updated with cryptogram+MAC, with the ACK itself used as key in security calculation;

4) Only one security file can be created in the same DF, with file name set as EF00, otherwise the card returns 6A80; when creating security file, the number of keys must be greater than equal to the default number of keys at the time of ADF/DDF creation; to update key, permit updating of existing keys prior to the creation of the security file;

5) Multiple counter files may be created within either DDF or ADF;

6) Under the same DDF, AID of all files include this DDF must be unique;

7) Under the same ADF, FID of all files include this ADF must be unique;

8) All data files (including binary files, record files, record files with extension field, counter files) could only be accessed through SFI; SFI valid range is 1~30, so under a DF there could be a maximum of 30 data files created;

9) When creating file, space must not be 0000;

10) Neither file record length nor number could be 00;

11) When creating files bundled with extension field and counter, see Appendix B for rules on bundled counter files.

——Command message (introduction of APDU encoding and parameters)

Table 9.3　　Create File command message

Code	**Value**
CLA	80/84
INS	E0
P1	00

Code	Value	
P2	01	Create ADF file
	02	Create binary file
	03	Create fixed length record file
	04	Create circular record file
	05	Create variable length record file
	06	Create counter file
	07	Create security file
	08	Create record file with extension field
	09	Create file bundled with extension field and counter
LC	Data length	
DATA	See "Create File data field description" (if CLA=84, then requires MAC to be included)	
LE	None	

——Create File data field description

- Data field upon creating ADF file

Table 9.4 Create File data fields (I)

File Type	Length (byte)	Notes
ADF	2	FID
	2	File size
	1	FCI-SFI (FCI file, must be binary file, includes the entire FCI data, if 00 then card returns the default FCI data - see Select File command description)
	1	ACK number of re-tries
	1	Property of ACK initial value: 0x0: 16 byte 0x00; 0x1: inherit the DACK in DDF; 0x2: inherit the DAAK in DDF;
	1	Application block/unblock key id, key type is application maintenance key or control key
	1	AID Tag, fixed as 0x4F
	1	AID length
	5-16	Application AID
	1	Application name Tag, fixed as 0x50
	1	Length of application name
	1~16	Application name (use easily understood names, can be displayed to the cardholder for selecting application)

- Data fields when creating elementary file (binary file, fixed length record file, circular record file, variable length record file, security file):

Table 9.5 Create File data fields (II)

File Type	**Length (byte)**	**Notes**	
Elementary file (EF)	2	FID, the lower 5 bits of the second byte is the SFI.	
	2	File size (all hexadecimal): 1) Binary file: 2 bytes representing the file size (space initially filled with 0x00); 2) Fixed length record file: 1 byte of record number and 1 byte of record length (record pre-set, with record space filled with 0x00); 3) Circular record file: 1 byte of record number and 1 byte of record length (record pre-set, with record space filled with 0x00); 4) Variable length record file: 2 bytes representing file size (length of each record determined by the first Append Record)	
	1	Read right	xy: required security level
	1	Write right	xy: required security level

File Type	Length (byte)	Notes	
	1	Read control property	Bit6=1, read operation requires encryption bit2=1, logical relationship between PIN right and read right. '0' = 'AND' relationship; '1' = 'OR' relationship bit1=1, read file requires PIN authentication, other bits are reserved
	1	Write control property	bit6=1, write operation requires encryption bit5=1, write operation requires MAC bit2: logical relationship between PIN right and write right. '0' = 'AND' relationship; '1' = 'OR' relationship bit1=1, write file requires PIN authentication, other bits are reserved
	1	Key id used for read and write control property	Key type is application maintenance key (id: 0x01~oxFE) or control key (id: 0x00)

- Create security file

Table 9.6 Create File data fields (III)

Security file	2	File id (fixed as 0xEF00)

	1	Number of keys

- Create counter file

Table 9.7 Create File data fields (IV)

Counter File	2	FID, the lower 5 bits of the second byte is the SFI.	
	1	Counter number	
	1	Increment (deposit) key ID	xx: counter increment key id, key type is INCK (id:0x56~0xFF) or control key (id: 0x00)
	1	Decrement (purchase) key ID	yy: counter decrement key id, key type is DECK (id:0x56~0xFF) or control key (id: 0x00)
	1	Increment/decrement control property	bit8=1, require PIN authentication prior to decrement bit7=1, require PIN authentication prior to increment Other values: reserved

- Create record file with extension field

Table 9.8 Create File data fields (V)

Record file with extension field	2	FID, the lower 5 bits of the second byte is the SFI.
	2	1 byte record number and 1 byte record length (max of 230 bytes, not including key length)

	1	Initial key value source: 00: Inherit application control key 01~0xFE: inherit application maintenance key

- Create file bundled with extension field and counter

Table 9.9 Create File data fields (VI)

File bundled with extension field and counter	2	FID, the lower 5 bits of the second byte is the SFI of the record file with extension field.	
	2	1 byte record number and 1 byte record length (max of 230 bytes, not including key length)	
	1	Initial key value source: 00: Inherit application control key 01~0xFE: inherit application maintenance key	
	1	Increment (deposit) key ID	Counter increment key id, key type is INCK (id:0x56~0xFF) or control key (id: 0x00)

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.10 Create File response message status

SW1	SW2	Meaning
90	00	Command executed successfully

SW1	SW2	Meaning
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	82	Security status not met
69	84	Random number not requested
6A	80	Data field parameter error
6A	84	Insufficient space
69	85	Usage conditions not met
65	81	Memory failure, non-volatile memory state has changed, command processing failed
94	03	Key not found

9.2.2 Delete DF Command

——Command description:

Delete DF command is used to delete ADF, after obtaining delete right in DDF, can delete the ADF in DDF, after obtaining delete right in ADF can delete the current ADF.

——Usage conditions and security:

- To delete the ADF in DDF requires verification of DDF control key (DACK);
- After ADF control key authentication can delete the current ADF;

——Command message (introduction of APDU encoding and parameters)

Table 9.11 Delete DF command message

Code	Value
CLA	80
INS	0E
P1	00: Delete the ADF in the current DDF directory
	01: Delete current ADF directory
P2	00
LC	Data length
DATA	If P1=00, data field is 5-16 byte ADF directory name If P1=01, no data entry is needed
LE	None

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.12 Delete DF response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error

SW1	SW2	Meaning
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	82	Security status not met
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.3 Read Count: Command to Read Counter Value

——Command description: Command is used to read the counter value in counter files

- For the counter file to be read, use 1 byte counter file id and 1 byte counter serial number to arrive at the unique counter position.
- Read Count command leverages counter serial number to read one counter value at a time.

——Usage conditions and security: No security restrictions on reading counters, can be freely read.

——Command message (introduction of APDU encoding and parameters)

Table 9.13 Read Count command message

Code	Value
CLA	80
INS	5C
P1	Counter serial number (0x01-0xFF)

Code	Value
P2	Counter file SFI (see table below)
LC	None
DATA	None
LE	04

P2 encoding:

Table 9.14 Read Count command P2 encoding

Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning
1	0	0						Access through SFI
			X	X	X	X	X	Counter file SFI

——Data in response

Table 9.15 Read Count response data

Notes	Length (byte)
Counter value	4

——Response message status (SW1, SW2)

Table 9.16 Read Count response message status

SW1	SW2	Meaning
90	00	Command executed successfully

SW1	SW2	Meaning
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	81	Command and file structure mismatch
6A	82	Counter file does not exist
6A	83	Counter does not exist
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.4 Increase Count: Command to Increase Counter Value

——Command description:

Increase Count command is used to increase the counter value of the designated counter.

- For the counter file to be increased, use 1 byte counter file id and 1 byte counter serial number to arrive at the unique counter position.

——Usage conditions and security: Based on counter file security definition, the data field format for this command is: 4 byte counter change increment + 4 byte MAC.

- Before executing this command, must obtain 4 byte random number using GET CHALLENGE command
- For MAC calculation method, see the corresponding content in Section 8.5.2.
- For MAC calculation key, use the increase count key to calculate MAC;

- After command is executed, new value of counter = original value of counter + counter change increment

——Command message (introduction of APDU encoding and parameters)

Table 9.17 Increase Count command message

Code	**Value**
CLA	84
INS	5D
P1	Counter serial number (0x01-0xFF)
P2	Counter file SFI (see table below)
LC	8
DATA	4 byte counter change increment + 4 byte MAC
LE	None

P2 encoding:

Table 9.18 Increase Count command P2 encoding

Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning
1	0	0						Access through SFI
			X	X	X	X	X	Counter file SFI

——Data in response

None

——Response message status (SW1, SW2)

Table 9.19 Increase Count response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
6A	82	Counter file does not exist
6A	83	Counter does not exist
69	81	Command and file structure mismatch
69	84	Random number not requested
69	88	MAC verification error
94	02	Counter overflow, post-transaction amount less than 0 or greater than 0xFFFFFFFF
65	81	Memory failure, non-volatile memory state has changed, command processing failed
94	03	Key not found

9.2.5 Decrease Count: Decrease Counter Value

——Command description:

Decrease Count command is used to decrease the counter value of the designated counter.

- For the counter file to be decreased, use 1 byte counter file id and 1 byte counter number to arrive at the unique counter position.

——Usage conditions and security:

- Based on counter file security definition, the data field format for this command is: 4 byte counter change increment + 4 byte MAC;
- Before executing this command, must obtain 4 byte random number using GET CHALLENGE command;
- For MAC calculation method, see the corresponding content in Section 8.5.2;
- For MAC calculation key, use the decrease count (purchase) key to calculate MAC;
- After command is executed, new value of counter = original value of counter - counter change increment

——Command message (introduction of APDU encoding and parameters)

Table 9.20 Decrease Count command message

Code	**Value**
CLA	84
INS	5E
P1	Counter serial number (0x01-0xFF)
P2	Counter file SFI
LC	8
DATA	4 byte counter change increment + 4 byte MAC
LE	None

P2 encoding:

Table 9.21 Decrease Count command P2 encoding

Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning
1	0	0						Access through SFI
			X	X	X	X	X	Counter file SFI

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.22 Decrease Count response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	81	Command and file structure mismatch
69	84	Random number not requested
6A	82	Counter file does not exist
6A	83	Counter does not exist
69	88	MAC verification error

SW1	SW2	Meaning
94	02	Counter overflow, post-transaction amount less than 0 or greater than 0xFFFFFFFF
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.6 Get Data: Obtain Profile Information from IC Card

——Command description:

The Get Data command obtains profile data from the IC card through Tag, such as common data store space status, etc.

——Usage conditions and security: No security requirements for this command, can be executed at will.

——Command message (introduction of APDU encoding and parameters)

Table 9.23 Get Data command message

Code	Value
CLA	80
INS	CA
P1	Tag high byte
P2	Tag low byte
LC	None
DATA	None
LE	Expected value length

——Data in response:

Table 9.24 Get Data response data

TAG(P1P2)	Name	Value (byte)	Content
DF 01	Common data store application version no.	2	For cards meeting this Specification, the value is fixed at 0100, or version 1.0
DF 02	Total space in current common data store region	2	Use byte as the unit, hexadecimal display of return, so 2000 means 8192 bytes, or 8KB.
DF 03	Remaining usable space in current common data store region	2	Use byte as the unit, hexadecimal display of return, so 1000 means 4096 bytes, or 4KB.

——Response message status (SW1, SW2)

Table 9.25 Get Data response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
69	01	Can not process

SW1	SW2	Meaning
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.7 Write Key: Command to Create/Update Key

——Command description: This command is used to create and update keys for common data store application.

——Usage conditions and security:

- Set up a new key, if the designated key already exists, Write Key command will indicate error and not execute;
- Update an existing key, if the designated key does not exist, Write Key command will indicate error and not execute;
- If key update is successful then key retries allowed will return to max value;
- For MAC and data encryption calculation method, see the corresponding content in Section 8.5.2;
- MAC and data encryption algorithm uses the application control key;
- Command message format is fixed as cryptogram+MAC.

——Command message (introduction of APDU encoding and parameters)

Table 9.26 Write Key command message

Code	Value
CLA	84
INS	D4

Code	Value
P1	00: Create key 01: Update key
P2	00
LC	Variable length
DATA	Cryptogram+MAC Therein: When P1=00, see table below for cryptogram pre-encryption format; key type 00 keys are automatically created during ADF creation, and do not require use of Write Key command. The two key type 00 keys (DACK, DAAK) in the common data store space DDF are simultaneously created during DDF creation, use default all zero hexadecimal as key value, and key retry times can be configured during DDF creation. When P1=01, cryptogram pre-encryption format is the cryptogram for "key type+key id+key value"
LE	None

Plaintext key format during key or PIN creation:

Table 9.27 Write Key command message key format

Bytes **Type**	**1**	**1**	**2**	**1**	**1**	**16**
Type 1	01	ID	0000	Limit	00	Key-Data
Type 2	02	ID	0000	Limit	ACL	Key-Data

Bytes Type	1	1	2	1	1	16
Type 3	03	ID	0000	00	00	Key-Data
Type 4	04	00	DPRK|| 00	Limit	00	PIN(8)

Therein:

1) Type: see description in 8.2 "Key Types"

2) Index: key reference serial number;

3) Limit: Key/PIN authentication limit. Number of consecutive authentication failure, max configuration of 15 times. When Limit drops to 0, key and PIN will be blocked. If Limit is set to FF, that means no limit;

4) ACL: Security grade obtained after passing authentication, only valid for external authentication PIN, format is 0x, x indicating privilege grade (between 0~F)

5) Key-Data: key data. 16 bytes;

6: PIN: password data, 8 bytes. Its valid length is 2~6 bytes. When creating PIN initially, PIN value is 6 bytes, composed of: PIN value = valid bytes+fill data 'FF' (6~2 bytes). When updating PIN only enter valid 2~6 bytes, without needing fill data 'FF';

Ex.: PIN value is '1234'

New PIN: PIN value = '1234FFFFFFFFFFFF' Update command: PIN value='1234'

7) DPRK: PIN reload key id, key type is application maintenance key.

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.28 Write Key response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	82	Security status not met
69	88	MAC verification error
69	84	Random number not requested
6A	84	Insufficient space in file
65	81	Memory failure, non-volatile memory state has changed, command processing failed
94	03	Key not found

9.2.8 Get Challenge: Command to Obtain Random Number

——Command description:

Random number command to request a 4 or 8 byte random number for use in security related process.

——Usage conditions and security: This random number can only be used for issuing one command, no matter whether the next command uses this random number, this random number will immediate expire.

——Command message (introduction of APDU encoding and parameters)

Table 9.29 Get Challenge command message

Code	Value
CLA	00
INS	84
P1	00
P2	00
LC	None
DATA	None
LE	04/08

——Data in response

Table 9.30 Get Challenge response data

Notes	Length (byte)
Random number	4 or 8

——Response message status (SW1, SW2)

Table 9.31 Get Challenge response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error

SW1	SW2	Meaning
6A	86	Parameter P1, P2 error
67	00	Length error
6A	81	This function is not supported

9.2.9 Internal Authenticate: Command for Internal Authentication

——Command description: Command provides the functionality to utilize the random number sent by interface device and self-stored relevant keys for data authentication. IC card encrypts the data provided by interface device, also returns the authentication data generated back to the interface device.

——Usage conditions and security:

The key that the command uses (designated by P2 parameter), must meet key access rights.

——Command message (introduction of APDU encoding and parameters)

Table 9.32 Internal Authenticate command message

Code	Value
CLA	00
INS	88
P1	See P1, P2 parameter table
P2	See P1, P2 parameter table
LC	Length of authentication data
DATA	Authentication data

Code	Value
LE	08

——P1, P2 parameter table

Table 9.33 Internal Authenticate parameter setup

	Bit8	**Bit7**	**Bit6**	**Bit5**	**Bit4**	**Bit3**	**Bit2**	**Bit1**	**Meaning**
P1	0	0	0	0	0	0	0	0	Use internal authentication key
	1	0	0	x	x	x	x	x	Use key in extended record file, lower 5 bits indicates file SFI
P2	X	X	-{}- X	X	X	X	X	X	When using internal authentication keys, this value represents internal authentication key index
	X	X	X	X	X	X	X	X	When using key in record file, this value represents record number

If using key for record file with extension as part of internal authentication, the key must be derived using 8 byte '00' prior to calculation.

——Data in response

Table 9.34 Internal Authenticate response data

Notes	Length (byte)
Authentication data	8

——Response message status (SW1, SW2)

Table 9.35 Internal Authenticate response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6A	83	Record does not exist
6A	86	Parameter P1, P2 error
67	00	Length error
69	81	File type error
94	03	Key not found

9.2.10 External Authenticate: Command for External Authentication

——Command description:

Goal of this command is to have IC card verify the validity of external interface device, allowing the interface device to obtain certain operating rights for IC card.

——Usage conditions and security: Key verification failure counter decrease by one. When counter decreases to '0' value, key will be blocked.

After choosing DF (regardless of whether choosing current DF or another DF), the security grade already obtained through external authentication command for the current DF will all expire.

——Command message (introduction of APDU encoding and parameters)

Table 9.36 External Authenticate command message

Code	Value
CLA	00
INS	82
P1	00
P2	XX: Key ID Therein: 00 : DACK, ACK 01～FE: DAEK, AEK FF : DAAK (only valid in DDF)
LC	Length of authentication data
DATA	Authentication data
LE	None

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.37 External Authenticate response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error

SW1	SW2	Meaning
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	83	Key is blocked
69	84	Random number is invalid
94	03	Key not found
63	CX	Authentication failure, X indicates retries remaining
64	00	Non-volatile memory state change, command processing complete but not correct

9.2.11 Read Binary: Command to Read Binary File

——Command description: This command is used to read the content of binary files.

——Usage conditions and security:

- Command execution must meet the read rights of file accessed;
- When using SFI to read the binary file in the DF currently designated by the file pointer, regardless of whether the read operation succeeds, the current file pointer position will not change;
- When the file read property is set as requiring encryption, LE must be 00 or a whole number multiple of 8 in length, if LE is not equal to 00, LE indicates the actual length of the cryptogram data. This cryptogram has at least LE-1 bytes of plaintext. If the length of remaining plaintext data is greater than LE-1, then length takes LE-1 bytes of plaintext data, if the length of remaining plaintext data is less than LE-1, then take all plaintext data. If the obtained plaintext data after encryption has cryptogram data length less than LE, then

return 6CXX. For example if LE equals 08, if remaining plaintext data is greater than 7 bytes, then card must read 7 bytes of data, following the format below to organize data:

07||7 bytes after reading data, encryption, then return to terminal. If the remaining plaintext data has less than 7 bytes, for example 3 bytes, then card must follow the format below to organize data: 03||read all data||80000000, after encryption return to terminal. Because a single data response value has a max of 255, if it is cryptogram data then its length would be a whole number multiple of 8, so the max cryptogram data response from Read Binary is 248 bytes. Therefore, it is necessary to consider length-related issues of relevant file properties when designing binary files. When file read property is set as requiring encryption, binary file length shall not exceed 502 bytes.

——Command message (introduction of APDU encoding and parameters)

Table 9.38 Read Binary command message

Code	**Value**
CLA	00
INS	B0
P1	See table 9.39
P2	See table 9.39
LC	N/A
DATA	N/A
LE	Data length of expected response

P1, P2 parameter table:

Table 9.39 Read Binary parameter configuration

P1	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning

	1	0	0	x	x	x	x	x	Access through SFI
P2	Initial offset								

——Data in response: For expected return data COS will decide based on the read control property setup at file creation time to return plaintext or cryptogram, with cryptogram being in multiple of 8 bytes.

——Response message status (SW1, SW2)

Table 9.40 Read Binary response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	81	Command and file structure mismatch
69	82	Security status not met
6A	82	File not found
6B	00	Offset error
69	88	Security message invalid
6C	XX	Le error, 'xx' indicates actual length

SW1	SW2	Meaning
61	XX	Normal processing, XX obtains additional data length through GET RESPONSE
64	00	Non-volatile memory state change

9.2.12 Update Binary: Command to Update Binary File

——Command description:

Command is used to update transparent file content.

——Usage conditions and security:

- Command execution must meet the write rights and control property of the file accessed;
- When using SFI to read the binary file in the DF currently designated by the file pointer, regardless of whether the read operation succeeds, the current file pointer position will not change.

——Command message (introduction of APDU encoding and parameters)

Table 9.41 Update Binary command message

Code	Value
CLA	00/04
INS	D6
P1	See table 9.42
P2	See table 9.42
LC	Length of subsequent data field

Code	**Value**
DATA	1) Data for update 2) Data for update + message authentication code (MAC) data element
LE	None

P1, P2 parameter table:

Table 9.42 Update Binary parameter configuration

P1	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning
	1	0	0	x	x	x	x	x	Access through SFI
P2	Initial offset								

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.43 Update Binary response message status

SW1	**SW2**	**Meaning**
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error

SW1	SW2	Meaning
67	00	Length error
94	03	Key not found
69	81	Command and file structure mismatch
69	82	Security status not met
69	84	Random number not requested
6B	00	Offset error
69	88	Security message invalid
65	81	Memory failure, non-volatile memory state has changed, command processing failed
94	03	Key not found

9.2.13 Append Record: Command to Add Record

——Command description: Command is used to append record to circular record file and linear variable length record file. For circular record file, when record is filled up, executing this command will overwrite the earliest written record, the last written record has record number 1 as the newest record; for linear variable length record file, this command can only add record and not over-write records, using Update Record command to modify the record.

——Usage conditions and security:

- Command execution must meet the write rights and control property of the file accessed;
- When using SFI to read the circular record file and linear variable length record file in the DF currently designated by the file pointer, regardless of whether the read operation succeeds, the current file pointer position will not change.

——Command message (introduction of APDU encoding and parameters)

Table 9.44 Append Record command message

Code	Value
CLA	00/04
INS	E2
P1	00
P2	See table 9.45
LC	Length of subsequent data field
DATA	Record data
LE	None

P2 parameter table

Table 9.45 Append Binary parameter configuration

P2	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning
	X	X	X	X	X	-	-	-	SFI

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.46 Append Binary response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	81	Command and file structure mismatch
69	82	Security status not met
69	84	Random number not requested
6A	80	Data field parameter error
6A	82	File not found
6A	84	Insufficient space in file
69	88	Security message invalid
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.14 Read Record: Command to Read Record

——Command description: Command is used to read record file content. This command is appropriate for reading linear fixed length file, circular file, linear variable length file, and records with extension field.

——Usage conditions and security:

- Command execution must meet the read rights of file accessed;
- When using SFI to read the circular record file and linear variable length record file in the DF currently designated by the file pointer, regardless of whether the read operation succeeds, the current file pointer position will not change;
- If the file read property is set to require encryption, then LE should be 00 or the length of actual cryptogram data after record data has been encrypted.

Because a single data response value has a max of 255, if it is cryptogram data then its length would be a whole number multiple of 8, so the max cryptogram data response from Read Record is 248 bytes. Therefore, it is necessary to consider length-related issues of relevant file properties when designing record files. When file read property is set as requiring encryption, each record length shall not exceed 247 bytes.

——Command message (introduction of APDU encoding and parameters)

Table 9.47 Read Record command message

Code	Value
CLA	00
INS	B2
P1	Record No. (0x01-0xFF)
P2	See table 9.48
LC	N/A
DATA	N/A
LE	Data length of expected response

P2 parameter table:

Table 9.48 Read Record parameter configuration

	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning
P2	X	X	X	X	X	1	0	0-	SFI

——Data in response: For expected return data COS will decide based on the read control property setup at file creation time to return plaintext or cryptogram, with cryptogram being in multiple of 8 bytes.

——Response message status (SW1, SW2)

Table 9.49 Read Record response message status

SW1	**SW2**	**Meaning**
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	81	Command and file structure mismatch
69	82	Security status not met
6A	82	File not found
6A	83	Record not found
69	88	Security message invalid

SW1	SW2	Meaning
61	XX	Normal processing, XX obtains additional data length through GET RESPONSE
64	00	Non-volatile memory state change, command processing complete but not correct

9.2.15 Update Record: Command to Update Record

——Command description: Command to update records of circular files, linear fixed length files, and linear variable length files.

——Usage conditions and security:

- Command execution must meet the write rights and control property of the file accessed;
- When using SFI to read the circular record file and linear variable length record file in the DF currently designated by the file pointer, regardless of whether the read operation succeeds, the current file pointer position will not change.

——Command message (introduction of APDU encoding and parameters)

Table 9.50 Update Record command message

Code	Value
CLA	00/04
INS	DC
P1	xx:: Designated record No. (0x01-0xFF)
P2	See table 9.51
LC	Data field length

Code	Value
DATA	1) Update record data 2) Data for update + message authentication code (MAC) data element
LE	None

P2 parameter table:

Table 9.51 Update Record parameter configuration

P2	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Meaning
	X	X	X	X	X	1	0	0-	SFI

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.52 Update Record response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error

SW1	SW2	Meaning
94	03	Key not found
69	81	Command and file structure mismatch
69	82	Security status not met
69	84	Random number not requested
6A	80	Data field parameter error
6A	82	File not found
6A	83	Record not found
6A	84	Insufficient space in file
69	88	Security message invalid
65	81	Memory failure, non-volatile memory state has changed, command processing failed
94	03	Key not found

9.2.16 Verify PIN: Command to Verify PIN

——Command description: Command is used to verify the accuracy of the PIN in command data field.

——Usage conditions and security:

For common data store applications, only DDF has PIN, all ADF in a DDF share a PIN, Verify PIN command can be executed either in DDF or ADF.

If PIN verification is successful, then security state register value is set as the key's follow-on state, while a PIN error allows the counter to be reset to initial value. If PIN verification error occurs, then PIN retry times is decreased by 1, while returning SW1 SW2 = 63CX. If PIN has been blocked, then the command cannot be re-executed, but can use RELOAD PIN command (for PIN) to unblock, causing the original PIN key record to resume normal, while if PIN error allows the counter to be reset to initial value. When card returns 63C0, it indicates that PIN cannot be retried, then if using VERIFY command again, then failure state code '6983' will be returned.

After PIN verification succeeds, choose the different ADF through Select command, its verification success state will not change; upon choosing DDF, PIN verification state will become invalid.

——Command message (introduction of APDU encoding and parameters)

Table 9.53 Verify PIN command message

Code	**Value**
CLA	00
INS	20
P1	00
P2	00
LC	02-06
DATA	Externally entered PIN
LE	None

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.54 Verify PIN response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	83	PIN is blocked
63	Cx	Authentication failure, X indicates retries remaining
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.17 Change PIN: Command to Change PIN

——Command description:

Command permits cardholder to change existing PIN to a new PIN. Valid PIN can be updated to default PIN, or update default PIN to a valid PIN.

After CHANGE PIN command is successfully completed, the card must undergo the following operations:

1) PIN retry counter is reset to the PIN retry upper limit;

2) Set original personal PIN to new personal PIN. The personal PIN (PIN) value in this command is transmitted in plaintext format. The PIN in command data is stored in 'cn' format, it does not require filling of complete bytes, only the lower half of the lowest valid byte may require filling with 'F'.

——Usage conditions and security:

1) Old PIN error, PIN error counter decrease by 1, return 63CX, when reduced to 0 the PIN is blocked;

2) If old PIN and new PIN data format are illegal (legal data format is 0-9 or F), return 6A80;

3) Execute CHANGE PIN after PIN is blocked, return 6983;

4) Change PIN command may be executed in DDF or ADF.

——Command message (introduction of APDU encoding and parameters)

Table 9.55 Change PIN command message

Code	**Value**
CLA	80
INS	5E
P1	01
P2	00
LC	05～0D
DATA	Current PIN || FF || New PIN
LE	None

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.56 Change PIN response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
65	81	Memory failure, non-volatile memory state has changed, command processing aborted
93	03	Application has been blocked
94	03	Key not found
69	85	Usage conditions not met
69	83	Key is blocked
6A	80	Data field parameter error
69	88	Security message invalid
63	CX	Authentication failure, X indicates retries remaining
64	00	Non-volatile memory state change, command processing complete but not correct

9.2.18 Reload PIN: Command to Reload PIN

——Command description:

RELOAD PIN command is used by issuer to generate a new PIN (may be same as original PIN) for the cardholder. After successfully executing RELOAD PIN command, IC card must complete the following operation:

- PIN error retry counter reset;
- IC card original PIN must be set to new PIN value.

——Usage conditions and security:

1) Reload PIN does not require authentication of original PIN value;

2) If old PIN and new PIN data format are illegal (legal data format is 0-9 or F), return 6A80;

3) When the number of MAC consecutive errors reach the max retry limit of the PIN reload key, then PIN reload key is blocked;

4) PIN data in command is transmitted in plaintext;

5) Reload PIN command may be executed in DDF or ADF.

6) Command message format includes MAC;

7) Key for MAC calculation is the PIN key header DPRK designated id key.

——Command message (introduction of APDU encoding and parameters)

Table 9.57 Reload PIN command message

Code	**Value**
CLA	80
INS	5E
P1	00
P2	00

Code	Value
LC	‘06’ ～ ‘0A’
DATA	Reloaded PIN and MAC
LE	None

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.58 Reload PIN response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
65	81	Memory failure, non-volatile memory state has changed, command processing aborted
94	03	Key not found
69	85	Usage conditions not met
69	83	Key is blocked

SW1	SW2	Meaning
6A	80	Data field parameter error
69	88	Security message invalid
63	CX	X indicates retries remaining
64	00	Non-volatile memory state change, command processing complete but not correct

9.2.19 Select File: Select File

——Command description:

Command is used to select file in IC card, while security files could not be selected.

——Usage conditions and security: None

——File search order:

P1＝04:

First determine whether to choose itself, then determine whether it is parent directory, if not found, then search for DF with matching file name in current DF. If not found, then search for matching DF amongst the sibling DFs, only limited to same level DFs, with cross-level selection not allowed.

——Command message (introduction of APDU encoding and parameters)

Table 9.59 Select File command message

Code	Value
CLA	00
INS	A4

Code	Value
P1	04: Choose application through DF name (only limited to DF files)
P2	00
LC	05-16 byte DF name (AID)
DATA	File ID, Fild name
LE	None

——Data in response:

FCI

——Response data structure:

DDF '6F L {84 L {DF name} A5 L {88 L {DIR-SFI}}}';

ADF: '6F L { 84 L {DF name} A5 L [{50 L {Application Tag}} {9F08 L {Common data store version number}}]'; the above is the default ADF FCI data, self-defined FCI data can be placed in template A5, with tag of 'BF05'.

——Response message status (SW1, SW2)

Table 9.60 Select File response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error

SW1	SW2	Meaning
67	00	Length error
93	03	Application has been blocked
6A	81	Function not supported, or no DDF in card, or card is blocked
6A	82	File not found
61	XX	Normal processing, XX obtains additional data length through GET RESPONSE

9.2.20 Get Response: Command to Get Response Data

——Command description:

Command provides a means of transferring APDU (or part of APDU) from card to interface device.

——Usage conditions and security: Only appropriate for T=0 protocol

If Le is 0 or smaller or equal to the max returnable data, then return Le bytes of data; if Le is greater than the maximum returnable data, return 6700.

——Command message (introduction of APDU encoding and parameters)

Table 9.61 Get Response command message

Code	Value
CLA	00
INS	C0
P1	00

Code	Value
P2	00
LC	None
DATA	None
LE	Expected data length

——Data in response:

Expected response data

——Response message status (SW1, SW2)

Table 9.62 Get Response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6A	86	Parameter P1, P2 error
6F	00	Data invalid
6C	XX	Le error, 'xx' indicates actual length
61	XX	Normal processing, XX obtains additional data length through GET RESPONSE

9.2.21 Append E-Record: Add Extension Record

——Command description: This command is use to add records with extension field (key data), including data and record maintenance key. Note: Reading records uses the Read Record command.

——Usage conditions and security:

- Command will first seek out empty record, then add new record in empty record;
- Use record initial key for security calculations to add record.

——Command message (introduction of APDU encoding and parameters)

Table 9.63 Append E-Record command message

Code	**Value**
CLA	84
INS	E2
P1	01
P2	See table 9.64
LC	Data field length
DATA	Updated record data + maintenance key cryptogram + MAC
LE	None

P2 parameter table:

Table 9.64 Append E-Record parameter configuration

	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Notes
P2	X	X	X	X	X	0	0	0-	File ID

——Data in response:

Record number of current APPEND success record

——Response message status (SW1, SW2)

Table 9.65 Append E-Record response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	81	Command and file structure mismatch
69	82	Security status not met
69	84	Random number not requested
6A	80	Data field parameter error
6A	82	File not found
6A	84	Insufficient space in file
69	88	Security message invalid
65	81	Memory failure, non-volatile memory state has changed, command processing failed
94	03	Key not found

9.2.22 Update E-Record: Update Extension Record

——Command description: This command is used to update records with extension field (maintenance key), can independently update data, or simultaneously update data and record maintenance key.

COS determines whether the command is pure update data or update data plus key based on length of Lc. If record data is equivalent to record length plus 4 (length of MAC), then update record data, if record length is equal to record length plus 28 (key cryptogram plus MAC), then update record and key simultaneously.

Note: Reading records uses the Read Record command.

——Usage conditions and security:

- Use the key of the record itself for security calculation of record update;

——Command message (introduction of APDU encoding and parameters)

Table 9.66 Update E-Record command message

Code	Value
CLA	84
INS	D8
P1	Record No.
P2	See table 9.67
LC	Data field length
DATA	1) Updated record data + MAC 2) Updated record data + maintenance key cryptogram + MAC data element Note: MAC and cryptogram calculation both use the maintenance key of the current record.

LE	None

P2 parameter table:

Table 9.67 Update E-Record parameter configuration

P2	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Notes
	X	X	X	X	X	0	0	0-	File ID

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.68 Update E-Record response message status

SW1	**SW2**	**Meaning**
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	81	Command and file structure mismatch
69	82	Security status not met

SW1	SW2	Meaning
69	84	Random number not requested
6A	80	Data field parameter error
6A	82	File not found
6A	84	Insufficient space in file
69	88	Security message invalid
65	81	Memory failure, non-volatile memory state has changed, command processing failed
94	03	Key not found

9.2.23 Delete E-Record: Delete Extension Record

——Command description: Command is used to delete records with the designated key property.

——Usage conditions and security:

- Command finds designated record based on record number;
- After verify PIN + control key external authentication, can delete any record;
- When using MAC to delete records, OS automatically verifies MAC accuracy using the corresponding record key, then deletes record;
- After record deletion all data in record is set to 0x00, key is restored to initial key (maintenance key designated at file creation time); at the same time COS internally sets a marker bit, indicating that this record is empty and ready for follow-on usage. After deletion the card enters into an idle state, upon direct reading of this record the card returns 6A83.

——Command message (introduction of APDU encoding and parameters)

Table 9.69 Delete E-Record command message

Code	Value
CLA	80/84
	80: Post PIN verification deletion of designated records 84: Post PIN verification deletion of designated records
INS	D7
P1	Record No. (0x01-0xFF)
P2	See table 9.70
LC	When CLA=80, LC=0; When CLA=84, LC=4;
DATA	When CLA=80, no data; When CLA=84, MAC
LE	None

P2 parameter table:

Table 9.70 Delete E-Record parameter configuration

	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Notes
P2	X	X	X	X	X	0	0	0-	File ID

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.71 Delete E-Record response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	82	Security status not met
69	84	Random number not requested
69	88	MAC error
6A	82	File not found
6A	83	Record not found
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.24 Search Record: Search Record

——Command description:

Search Record command is used to search record files (including fixed length records files, variable length record files, record files with extension field, but not including circular record files) for record number of record matching designated data.

——Usage conditions and security: Command execution must meet the read rights of the file accessed; regardless of whether the command operation is successful or not, the current card state does not change.

——Command message (introduction of APDU encoding and parameters)

Table 9.72 Search Record command message

Code	Value
CLA	00
INS	D9
P1	Search data initial offset in record (range: 0x00~0xFF)
P2	See table 9.73
LC	0x01~0x20
DATA	Search data
LE	None

Note: Only records where initial offset consecutive data and the search data completely match will be returned.

P2 parameter table:

Table 9.73 Search Record parameter configuration

P2	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Notes
	X	X	X	X	X	0	0	-0	File ID

——Data in response:

Return the record number from search, each record number is 1 byte long, return in order. If search returns 3 records with record numbers 01, 05, and 1A, then return 01051A. If no record numbers match conditions, return 6A83.

——Response message status (SW1, SW2)

Table 9.74 Search Record response message status

SW1	SW2	Meaning
61	XX	Normal processing, XX obtains additional data length through GET RESPONSE
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
69	81	Command and file structure mismatch
69	82	Security status not met
6A	82	File not found
6A	83	Record not found
64	00	Non-volatile memory state change, command processing complete but not correct
69	84	Random number not requested

SW1	SW2	Meaning
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.25 Application Block: Block Application

——Command description:

After the APPLICATION BLOCK command is successfully executed, block the current valid application (referring ADF). Blocked application can still be selected (after successfully selecting the application, return SW1_SW2='6A81'), but the files of the blocked application could not be accessed, any attempt to access the forms will get a return of SW1SW2='6A81'. In touch interface, blocked application can obtain FCI data from the Get Response command.

——Usage conditions and security:

APPLICATION BLOCK command execution utilizes check mode. Key used to calculate verification code creates ADF files at the designated application block/decryption key id.

Blocked ADF can only see Get Data, Get Challenge, Get Response, and Application Unblock, other related file operation return "0x6481", other commands return 6E00 or 6D00.

——Command message (introduction of APDU encoding and parameters)

Table 9.75 Application Block command message

Code	Value
CLA	84
INS	1E
P1	00
P2	00

Code	Value
LC	04
DATA	4 byte MAC
LE	None

——Data in response:

None

——Response message status (SW1, SW2)

Table 9.76 Application Block response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	84	Random number not requested
69	88	MAC error
65	81	Memory failure, non-volatile memory state has changed, command processing failed

9.2.26 Application Unblock: Unblock Application

——Command description:

After the Application Unblock command executes successfully, unblocks the previously blocked applications.

——Usage conditions and security:

Application unblock command execution utilizes check mode. Key used to calculate verification code creates ADF files at the designated application block/decryption key id. To prevent external risk, this command can be exercised without limit.

——Command message (introduction of APDU encoding and parameters)

Table 9.77 Application Unblock command message

Code	Value
CLA	84
INS	18
P1	00
P2	00
LC	04
DATA	4 byte MAC
LE	None

——Data in response:

None

——Response message status (SW1, SW2) for application not locked, the status word this command returns is for conditions not met (SW1, SW2="6985").

Table 9.78 Application Unblock response message status

SW1	SW2	Meaning
90	00	Command executed successfully
6E	00	CLA error
6D	00	INS error
6A	86	Parameter P1, P2 error
67	00	Length error
94	03	Key not found
69	84	Random number not requested
69	88	MAC error
65	81	Memory failure, non-volatile memory state has changed, command processing failed
69	85	Usage conditions not met

10 Main Flow for Application Implementation and Command Operation

10.1 Fast Common Application (Common Application) Directory Add Flow

10.1.1 Fast Common Application (Common Application) Directory Add

In the common data store space, issuer may have pre-set some fast common application directory or common application directory at the time of card issuance, after issuance, based on market requirements, may add new application directories, with the following operating procedure:

Terminal creates fast common directory or common directory ADF and the various application files within ADF through the Create File command within the common data store space.

1. Create application directory ADF

a) Terminal must first obtain the create right for the card, so through external authentication, verify card common data store space control key (DACK) or common data store space application add key (DAAK).

b) The Create File command indicates whether the newly created common application directory ADF initial control key is all 0 or is the control key inherited from the common data store space DDF.

c) If security message is required, Create File command also specifies whether security message is calculated with common data store space DDF control key (DACK) or with the common data store space DDF application add key (DAAK).

d) Calculation of external authentication cryptogram and security messages can be realized through terminal equipment (with PSAM card) or remote authentication with issuer back-end.

i. Terminal equipment authentication method: Terminal is equipped with PSAM card, with industry root key stored in PSAM card, terminal. Terminal uses PSAM card internal derived card sub-key to calculate authentication cryptogram and Create File command security message. This method generally uses DAAK.

ii. Issuer backend authentication method: Root key is stored in issuer back-end, terminal does not hold any key data. Terminal initiates request to issuer back-end, and issuer back-end calculates external authentication cryptogram and Create File command security message. This method generally uses DACK.

2. Create security file

a) In the create process and application directory ADF creation process, FID must be EF00;

b) In the create process, external authentication key and security message calculation key is the common application directory control key (ACK). Based on application directory create command configuration, this control key can be all 0 or the common data store space control key;

c) Under the protection of the application directory control key (ACK), through the Write Key command, write the various keys into the security file.

3. Create the various application files

a) Create process and security file;

b) Through external authentication command, terminal obtains the update right to application files;

c) For variable length record files and circular files, can use Append Record command to add records to file;

d) For all records files, use Update Record command to update the existing records in the file;

e) For transparent files, use Update Binary command to update file content.

10.1.2 Command Flow Chart

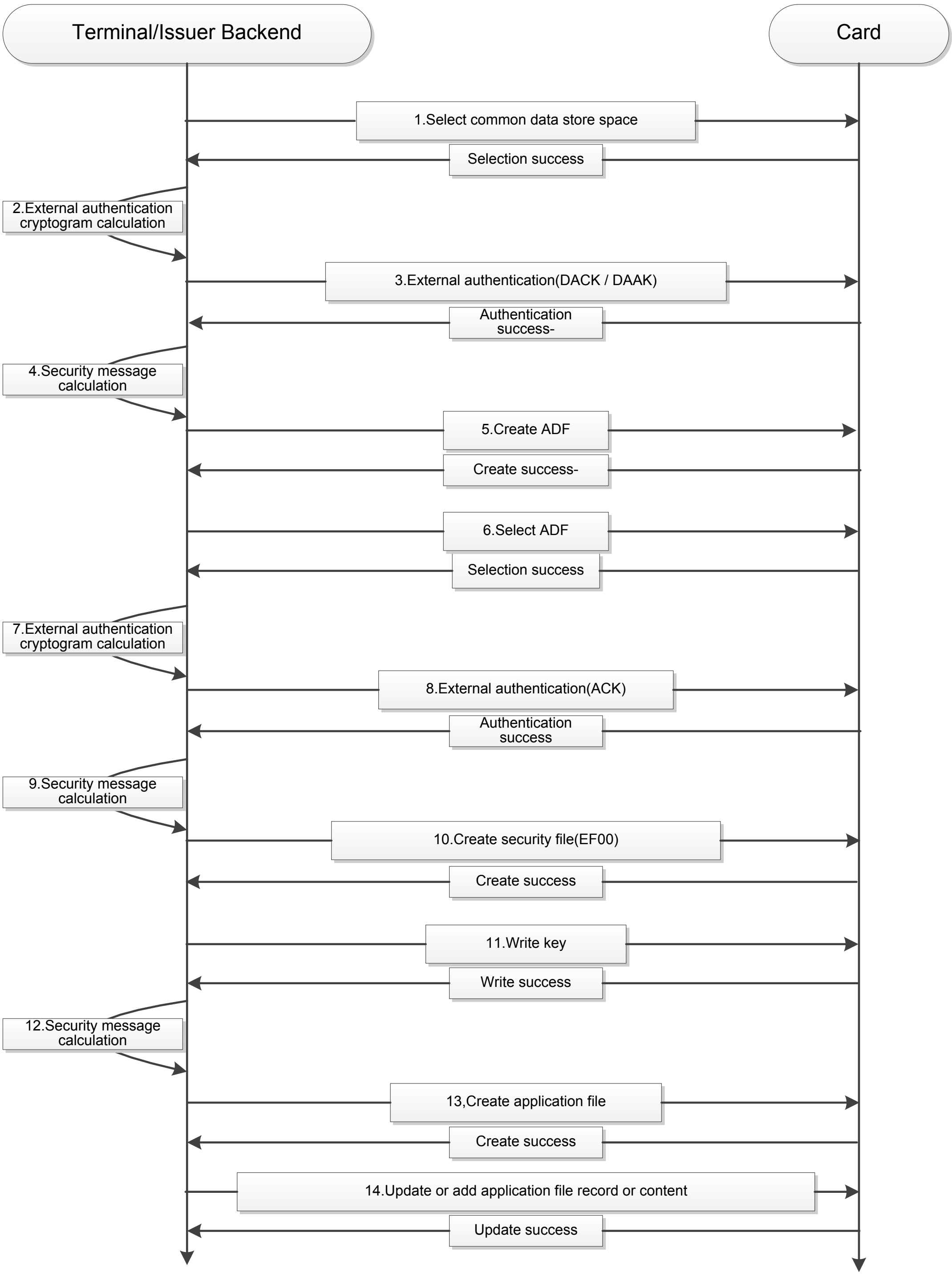


Figure 10.1. Fast common application (common application) direcotry add command flow chart

10.1.3 Command Flow Explanation

1. Select common data store space DDF

2. Through PSAM card or issuer backend calculate external authentication cryptogram, use DAAK or DACK

3. Execute external authentication, obtain file create right

4. Through PSAM card or issuer backend calculate create ADF file command security message (this step is optional)

5. Execute create file command to create application directory, create command specifies control key for newly created directory

6. Select newly created application directory

7. Through PSAM card or backend calculate external authentication cryptogram, use ACK

8. Execute external authentication, obtain file create right

9. Through PSAM card or issuer backend calculate create security file command security message (this step is optional)

10. Create security file, set FID to EF00

11. Use Write Key command to update control key, and add various maintenance keys and internal/external authentication keys

12. Through PSAM card or issuer backend calculate create application file command security message (this step is optional)

13. Create application files, including transparent files, variable length record files, fixed length record files, and circular files

14. Use Update Record, Append Record, and Update Binary command to update relevant files.

10.2 Flow to Add and Update Records to Fixed Length Record Files with Extension Field

10.2.1 Basic Concept of Files with Extension Field

Fixed length record files with extension field is a special kind of record that has individual control rights. This type of file as a different process for adding or updating of records from that of normal fixed length record files.

10.2.2 Add Record

Add record means adding industry application record to existing fixed length record file with extension field, with the add flow as follows: a) Adding record uses Append E-Record command to add new record content to the fixed length record file with extension field;

b) Based on the maintenance key ID designated at the creation of the fixed length record file with extension field, search for corresponding maintenance key and use it to calculate the security message for the Append E-Record command;

c) Card searches for empty records in the fixed length record file with extension field, after authenticating security message, writes the new record content into the empty record.

10.2.3 Command Flow Chart

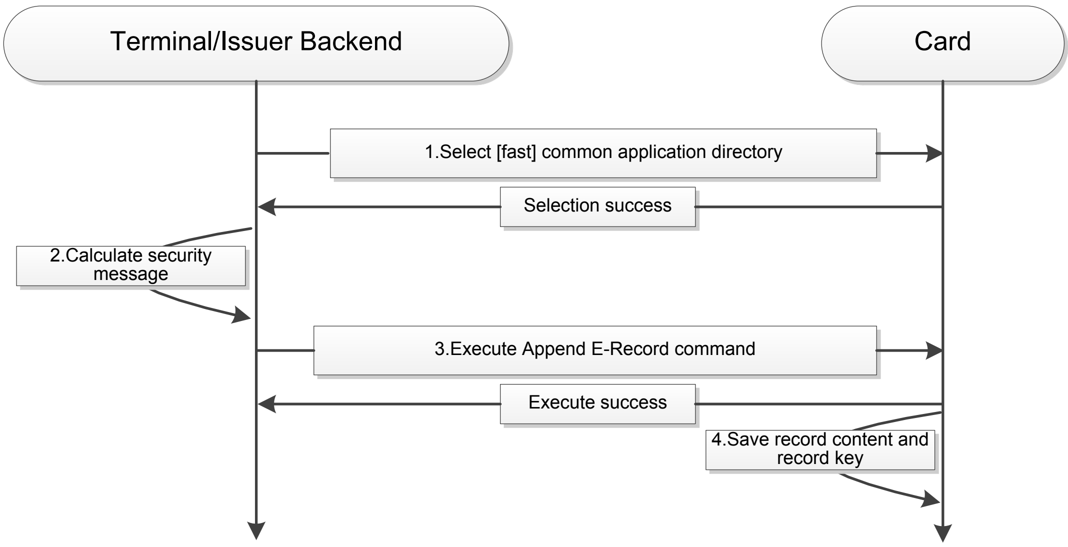


Figure 10.2 Flow to Add Record command for fixed length record files with extension field

10.2.4 Command Flow Explanation

1. Select fast common application directory or common application directory;

2. Use the initial maintenance key for the fixed length files with extension field to calculate command security message;

3. Execute Append E-Record command;

4. Save the record's own maintenance key included in the command, and save record content.

10.2.5 Update Record

Update record means updating the industry application record in existing fixed length record file with extension field, with the update flow as follows: a) Update record uses Update E-Record command to update record content in the fixed length record file with extension field;

b) Use record's own maintenance key to calculate Update E-Record command security message;

c) Card searches for the record specified in the command, and updates the new record content. If the command includes a new maintenance key, then the card uses to new maintenance key to replace the old maintenance key.

10.2.6 Command Flow Chart

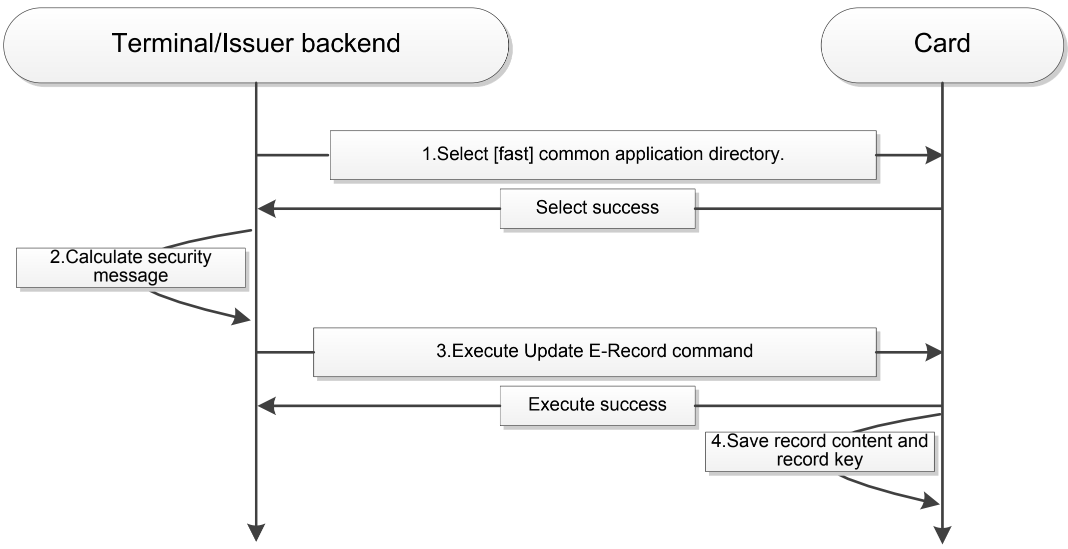


Figure 10.3 Flow to Update Record command for fixed length record files with extension field

10.2.7 Command Flow Explanation

1. Select fast common application directory or common application directory;

b) Use record's own maintenance key to calculate command security message;

3. Execute Update E-Record command;

4. Update the corresponding record content. If the command includes new record maintenance key, then also update record maintenance key.

10.3 Delete Application Directory Record

10.3.1 Delete Application Directory

The industry application in the common application directory can be, at the cardholder's request or in other circumstances, be deleted using the Delete DF command. Once the common application directory is deleted, it could not be recovered, and no application data can be regenerated. Common application directory can only be deleted by the issuer, while common application industry applications can be deleted by either the issuer or the corresponding industry. There are two types of common application directory means of deletion:

a) Issuer deletes industry application;

b) Industry deletes its own industry application.

10.3.2 Command Flow Chart

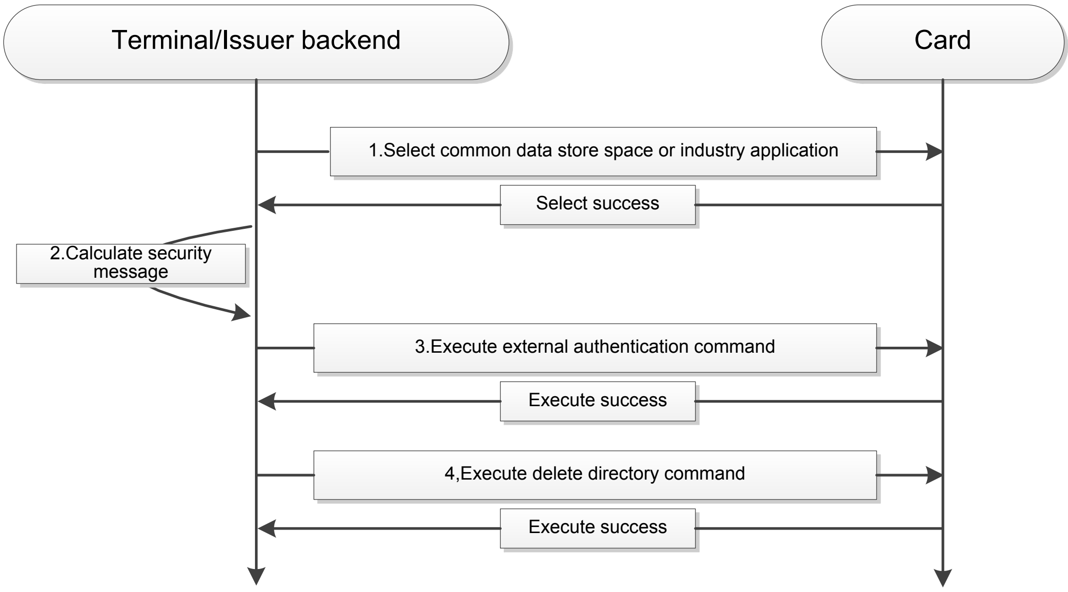


Figure 10.4 Application directory record deletion command flow chart

10.3.3 Command Flow Explanation

1. Based on deletion method, select common data store space or [fast] common application directory;

2. Based on deletion method, use DACK or ACK to calculate external authentication cryptogram;

3. Execute external authentication command, obtain delete right;

4. Execute delete directory operation.

10.4 Flow to Delete Records from Fixed Length Record Files with Extension Field

10.4.1 Deletion Flow for Record Files with Extension Fields Description

When industry application data stored in fixed length record file with extension field expires after usage, or the cardholder proactively gives up, the system provides a means of application deletion.

Record deletion can be initiated by cardholder, and can also be initiated by industry.

10.4.2 Record Deletion Initiated by Cardholder

Cardholder initiated record deletion operation must pass cardholder offline PIN verification, and be executed with issuer authorization.

10.4.3 Command Flow Chart

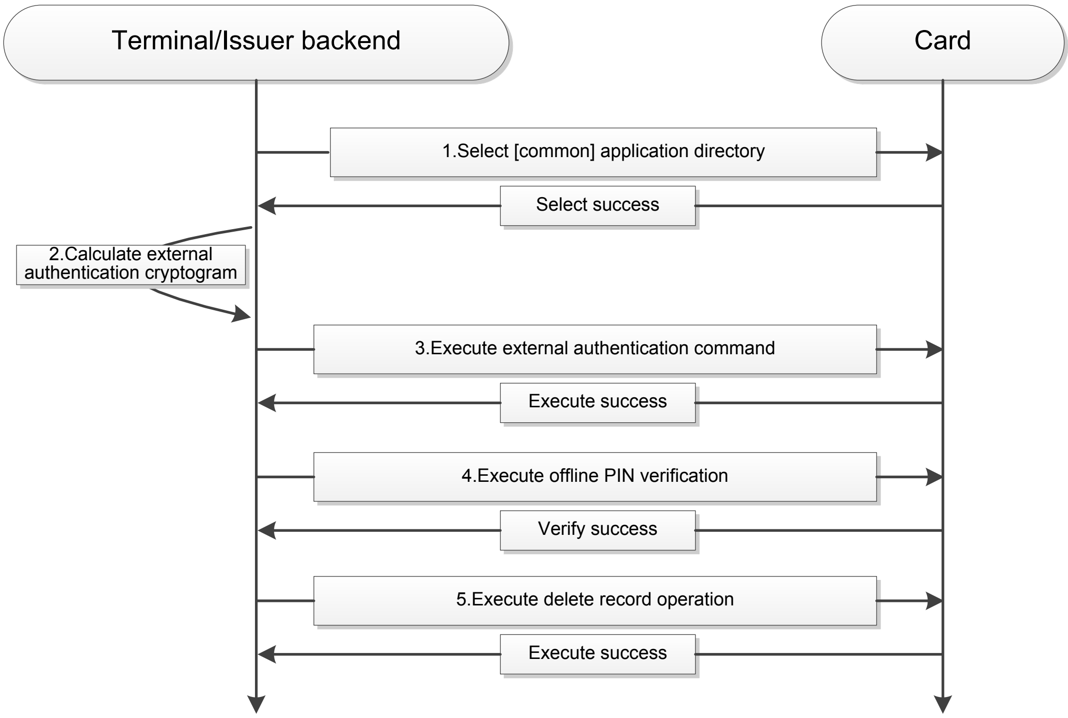


Figure 10.5 Cardholder initiated record delete command flow chart

10.4.4 Command Flow Explanation

1. Select [common] application directory;

2. Use ACK to calculate external authentication cryptogram;

3. Execute external authentication command, obtain issuer delete right;

4. Execute offline PIN verification, obtain cardholder agreement;

5. Execute Delete E-Record command to delete corresponding record.

10.4.5 Industry Initiated Record Deletion

Cardholder initiated record delete operation must obtain record key authentication before executing.

10.4.6 Command Flow Chart

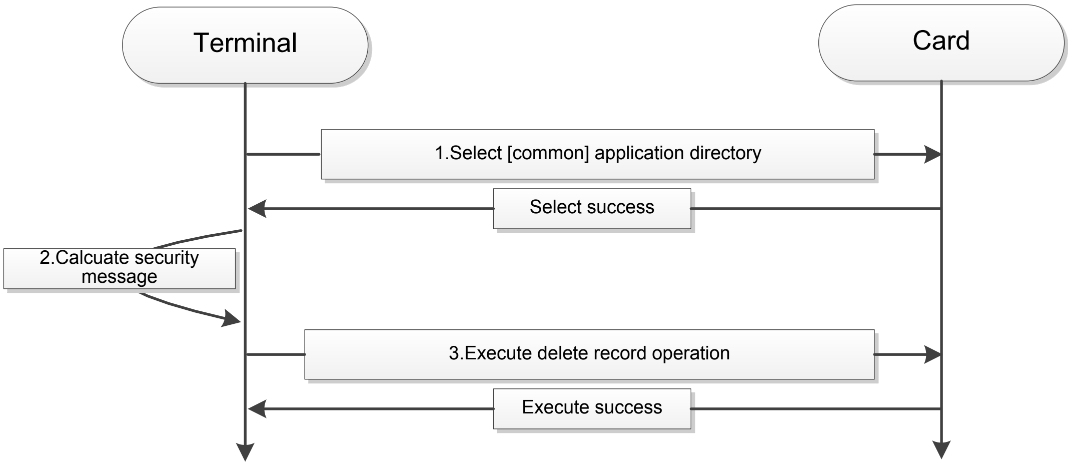


Figure 10.6 Industry initiated record delete command flow chart

10.4.7 Command Flow Explanation

1. Select common application directory;

2. Use record's own maintenance key to calculate delete record command security message;

3. Execute Delete E-Record command to delete corresponding record.

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Appendix A (Normative Annex) Fast Common Application Data File Implementation Requirements

This Section defines the basic structure of discount vouchers, e-tickets, and membership cards. Cards that meet this Specification must follow the corresponding structure below and requirements for file personalization.

A.1 Discount Voucher

A.1.1 Discount Voucher Basic Requirements

The discounter vouchers in this Specification uses smart cards as the carrier for digital information, and through terminal telecommunication capability to realize the write, modify, purchase, and deletion of discount vouchers and other functionality of traditional physical discount vouchers. Discount voucher application is a kind of application designed for cardholder to engage in discount voucher transactions. The discount voucher defined in this application refers to a single- or multi- use digital certificate, merchant follows the certificate defined discount rules to give corresponding discounts to users on purchases.

A.1.2 Discount Voucher File Definitions

Discount voucher is one of the basic functionality for fast common application directories. Relevant support application elementary files can be pre-set in fast common application directories. In fast common directories, there are two discount voucher information data files, one is a discount voucher data file that can be freely accessed, while the other is a discount voucher data file that includes key protected extension field, as shown below:

1. Discount voucher information data file Discount voucher information data file 1

Table A.1 Discount voucher information data file 1

File ID (SFI)	‘01h’
File ID (FID)	EF01
File Type	Fixed Length Record File
File size	'147*10' decimal

File Access Control		Read = FREE	Write = FREE
Field	Date element	Format	Length
1	Merchant Code	an	15
2	Discount Voucher Provider Code	ans	8
3	Discount Voucher Number	an	10
4	Discount Voucher Name	ans	20
5	Discount Voucher Information Template Type	B	1
6	Discount Voucher Information (for specific data format see template definition)		93

Discount voucher information data file 2

Table A.2 Discount voucher information data file 2

File ID (SFI)		‘02h’
File ID (FID)		EF02
File Type		Fixed Length Record File with Extension Field
File size		'147*10' decimal
File Access Control	Read = FREE	Write = maintenance key

Field	Date element	Form at	Length
1	Merchant Code	an	15
2	Discount Voucher Provider Code	ans	8
3	Discount Voucher Number	an	10
4	Discount Voucher Name	ans	20
5	Discount Voucher Information Template Type	B	1
6	Discount Voucher Information (for specific data format see template definition)		93

Discount voucher information template type: Merchant chooses appropriate information template record discount voucher information based on specific business needs and discount voucher type.

01: Pre-defined discount voucher

02: Merchant self-defined discount voucher Other: Reserved

01: Pre-defined discount voucher

Table A.3 Pre-defined discount voucher template

Field	**Date element**	**Length**	**Format**	**Code Description**
1	Discount voucher property	1	B	See table below
2	Text explanatory info	41	ans	Merchant defined simple explanatory info for this discount voucher, can be used for display at POS or terminal.

Field	Date element	Length	Format	Code Description
3	Discount voucher grouping	1	B	Merchant defined discount voucher grouping, merchant POS can leverage grouping to run batch operations on discount vouchers.
4	Discount rate	1	n	XX indicates can enjoy XX% discount.
5	Cash equivalent amount	4	n	Unit is cent, XXXXXX.XX represents cash equivalent amount of RMB XXXXXX.XX.
6	# vouchers/ # times	1	n	Indicates remaining usable times.
7	Validity Period (start)	4	n	YYYYMMDD, all 0 indicates this field is not used.
8	Validity Period (end)	4	n	YYYYMMDD, all 0 indicates this field is not used.
9	Min transaction amount	4	n	Unit is cent, XXXXXX.XX indicates that when purchase amount is greater than or equal to RMB XXXXXX.XX then discount voucher usage condition is met. Using all 0 indicates that there is no need to determine min transaction amount.
10	Max transaction amount	4	n	Unit is cent, XXXXXX.XX indicates that when purchase amount is lower than or equal to RMB XXXXXX.XX then discount voucher usage condition is met. Using all 0 indicates that there is no need to determine max transaction amount.

Field	Date element	Length	Format	Code Description
11	Registered info	20	an	Merchant self-defined record of user personal info for registered discount voucher.
12	Verification data	8	an	Merchant self-defined field, through verifying this data the discount voucher validity can be verified.

Various field descriptions:

Discount voucher property: 1 byte, with each bit representing a property

b7 b6 b5 b4 b3 b2 b1 b0 b0: 1 uses discount, 0 no discount b1: 1 cash equivalent amount, 0 non-cash equivalent amount b2: 1 count usage times, 0 do not count usage times

b3: 1 record name, 0 do not record name

b4: 1 check validity date, 0 do not check validity date Others are reserved

02: Merchant self-defined discount voucher

Table A.4 Self-defined discount voucher format

Field	Data	Format	Length
1	Text explanatory info length	n	1
2	Text explanatory info	an	
3	Discount voucher content length	n	1
4	Discount voucher content	an	

A.1.3 Card Personalization

Discount voucher application id (AID), application tag is set as A0 00 00 03 33 CD D0 01, and selects this application through application tag. Please see Section 10.1 for create method, with key steps as below.

Obtain random number and execute external authentication, obtain the discount voucher application create right to the main directory. (get challenge) creates discount voucher directory file through create file command, sets the access right (create DFOM), choose discount voucher application (select), create discount voucher application key file, discount voucher data information file, and point accumulation data file (create file). Write corresponding key (write key), update application file (update record, update binary), for specific format see APDU command explanation. Command Interaction Flow Chart:

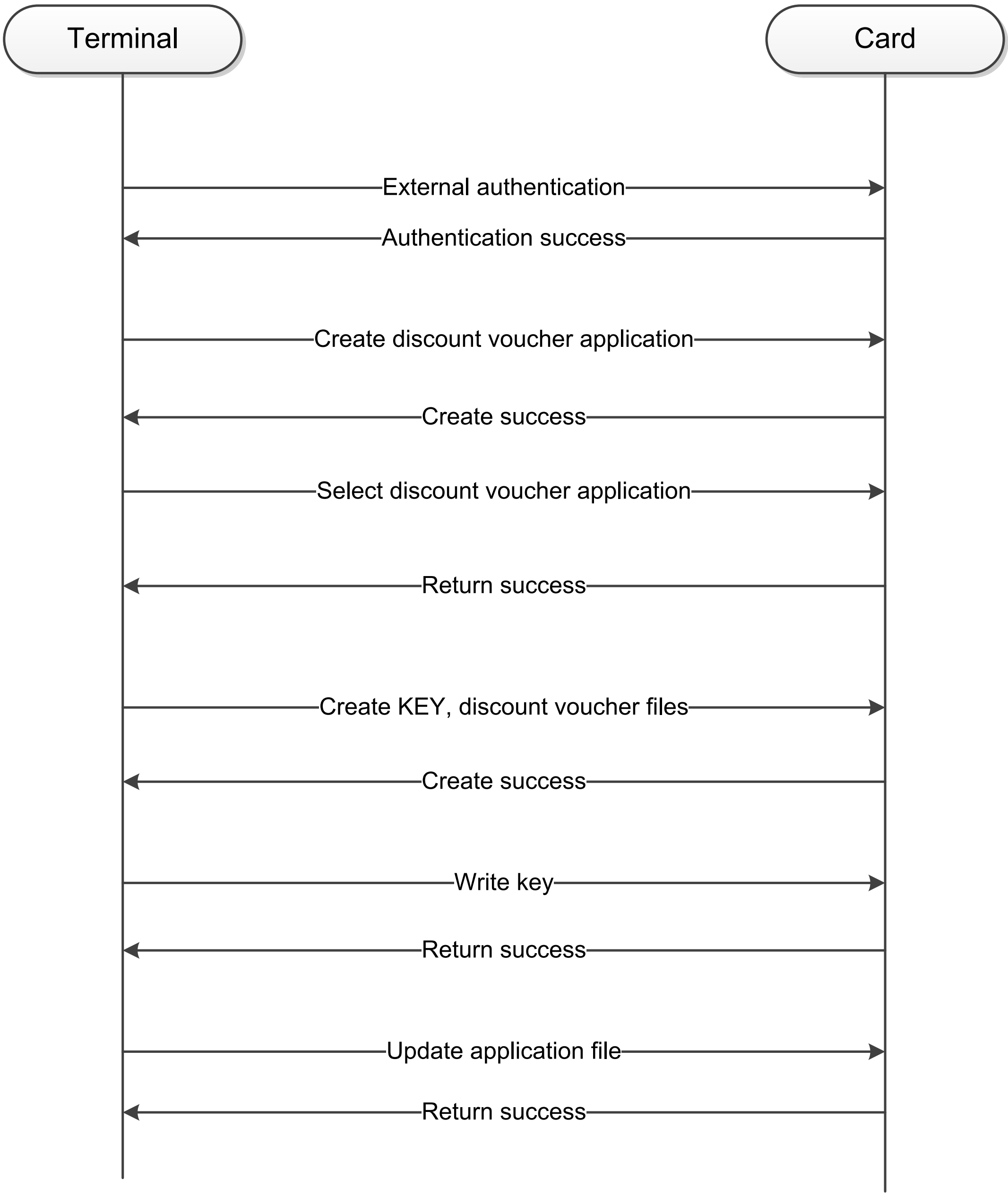

Figure A.1 Card personalization command interaction flow chart

A.1.4 Discount voucher business functionality

1. Discount voucher issuance Discount voucher issuance refers to the discount voucher provider or merchant writing the discount voucher info into user cards. Discount voucher provider or merchant can choose to write the discount voucher info into EF01 or EF02. EF01 stores low security required discount voucher types, with open read-write access. EF02 stores discount voucher that is protected by issuer-set key, protecting this discount voucher info from third-party revision or deletion.

Discount voucher main venues for issuance include:

- Centralized issuance point set up by the issuer or discount voucher provider: can search and download discount vouchers from multiple merchants

- Merchant: download merchant specific discount vouchers from merchant POS

- Personal internet terminal: Through remote connection discount voucher platform perform self-service search and download discount vouchers

2. Discount voucher purchase

User makes purchases from discount voucher participant merchants with card, with merchant terminal reading discount voucher info, and based on different discount voucher type and actual business requirements undergo follow-on operations.

- For discount vouchers with validity periods, first check whether it is within the validity period.

- For registered discount vouchers, first check the black list, and authenticate user identity as needed.

- For discount vouchers that track usage times, after normal purchase shall decrease the number of times/vouchers by one.

- For money type discount vouchers (cash equivalent vouchers), after normal purchase should delete this discount voucher.

3. Delete discount voucher

Deletion of discount voucher shall meet the 10.3 requirements for common application directory record deletion. Can be initiated by cardholder or discount voucher provider.

A.1.5 Discount voucher usage example

Operating flow and terminal -- card command flow example:

1. Discount voucher issuance

Discount voucher issuance means adding a new piece of discount voucher data in the discount voucher info file. The operating flow is as follows:

(1) Card holder inserts card;

(2) Verify cardholder offline PIN (or other right authentication method), obtain the right to write records;

(3) Follow the pre-determined format of the discount voucher info file to write record into card. For discount vouchers with low security grade requirement, no security protection is necessary when adding, write ordinary discount voucher files in directly. For discount vouchers with relatively high security grade requirements, can write key info into extended record file.

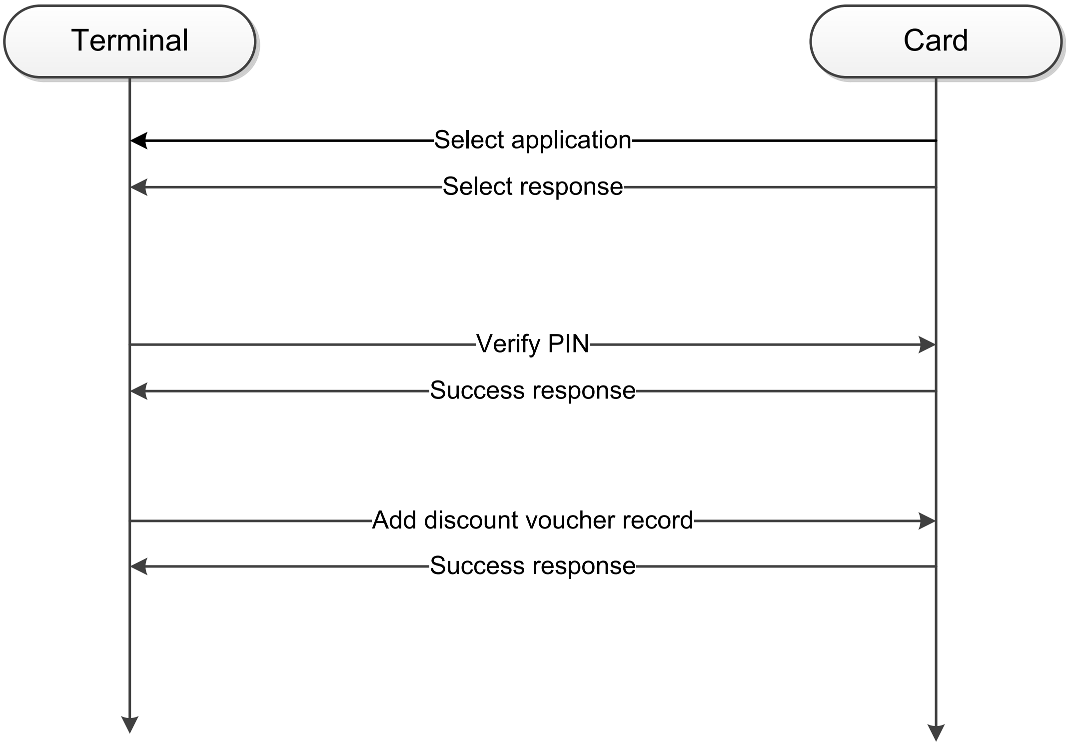

Figure A.2 Discount voucher issuance operating flow chart

Command Flow Explanation

a) Select discount voucher application (select);

b) PIN verification, adding discount voucher record requires cardholder agreement (verify pin);

c) Add record command, record can be added after passing cardholder verification. Based on security requirement, add record command can include the record's own maintenance key (update record update e-record).

d) Add record complete, if there is any corresponding key, insert key file, and establish corresponding relationship with record.

2. Discount voucher purchase

Discount voucher purchase is the operation for merchant to read discount voucher info and verify revision of discount voucher information. The operating flow is as follows:

(1) Card holder inserts card;

(2) Verify cardholder offline PIN (or other right authentication method), obtain the right to read records;

(3) Terminal reads all the discount voucher info belonging to the merchant, and the user designates the discount voucher to be used for purchase;

(4) Merchant applies discount to user purchase based on discount voucher content, based on different discount voucher type execute revision operation on discount voucher. If the discount voucher is valid for only 1 use, then the discount voucher shall be deleted after purchase completion; if valid for multiple uses, then after purchase completion the number of valid uses shall be decreased by 1 and updated to discount voucher info.

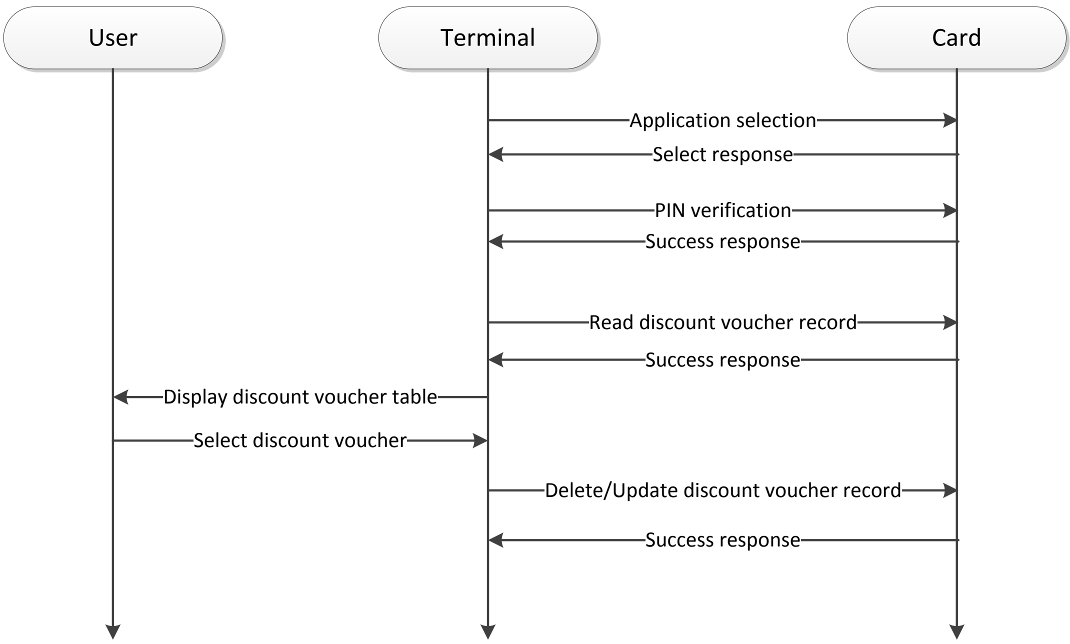

Figure A.3 Discount voucher purchase operating flow chart

Command Flow Explanation

a) Select discount voucher application (select);

b) PIN verification, discount voucher purchasing requires cardholder agreement (verify pin);

c) Read record command, reads all the discount voucher info table belonging to the merchant (read record, read e-record);

d) Purchase complete, based on discount voucher type execute corresponding delete or update operation on discount voucher info(update record, update e-record, delete e-record).

3. Delete discount voucher

When the discount voucher data store space is full, user can select to delete some discount vouchers. The operating flow is as follows:

(1) Card holder inserts card;

(2) Verify cardholder offline PIN (or other right authentication method), obtain the right to delete records;

(3) Terminal reads all the discount voucher info belonging to the merchant, and the user designates the discount voucher to be deleted;

(4) Terminal executes delete operation on the discount vouchers.

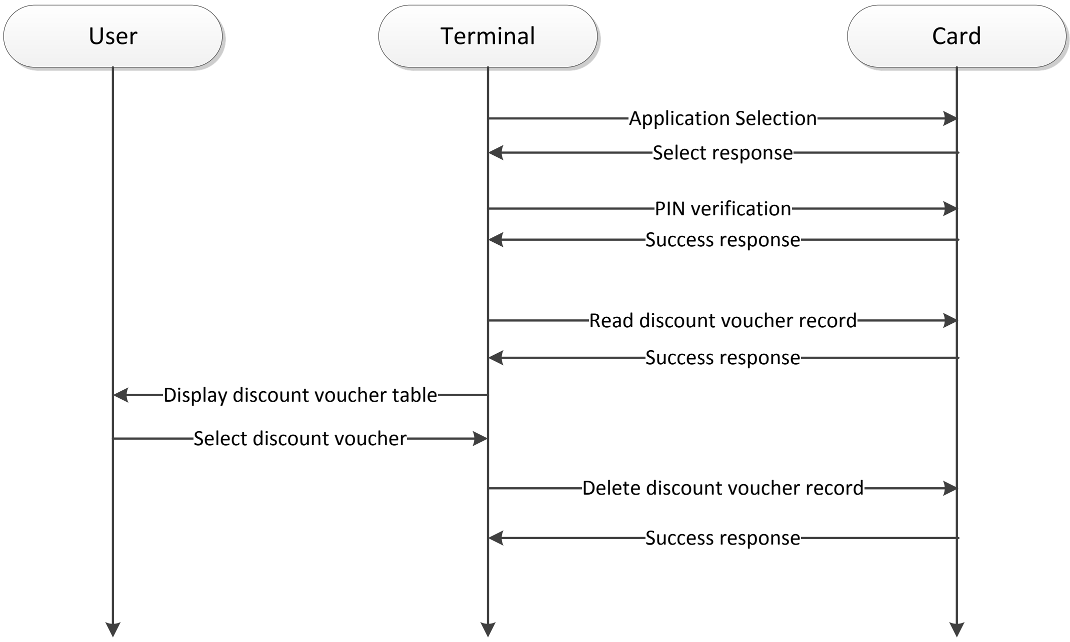

Figure A.4 Discount voucher delete operating flow chart

Command Flow Explanation

a) Select discount voucher application (select);

b) PIN verification, delete discount voucher record requires cardholder agreement (verify pin);

c) Read record command, reads all the discount voucher info table belonging to the merchant (read record, read e-record);

d) Based on user selected discount vouchers execute corresponding delete operation (update record, delete e-record).

A.2 E-ticket

A.2.1 E-ticket Business Functionality

A.2.1.1 E-ticket Introduction

Since IC card e-ticket application stores paper ticket info into IC cards, it can be used just like paper tickets on merchant terminals.

A.2.1.2 E-ticket Lifecycle

E-ticket lifecycle refers to the entire process from e-ticket data generation to e-ticket expiration, with the entire process divided into several traditional stages. As shown in figure below:

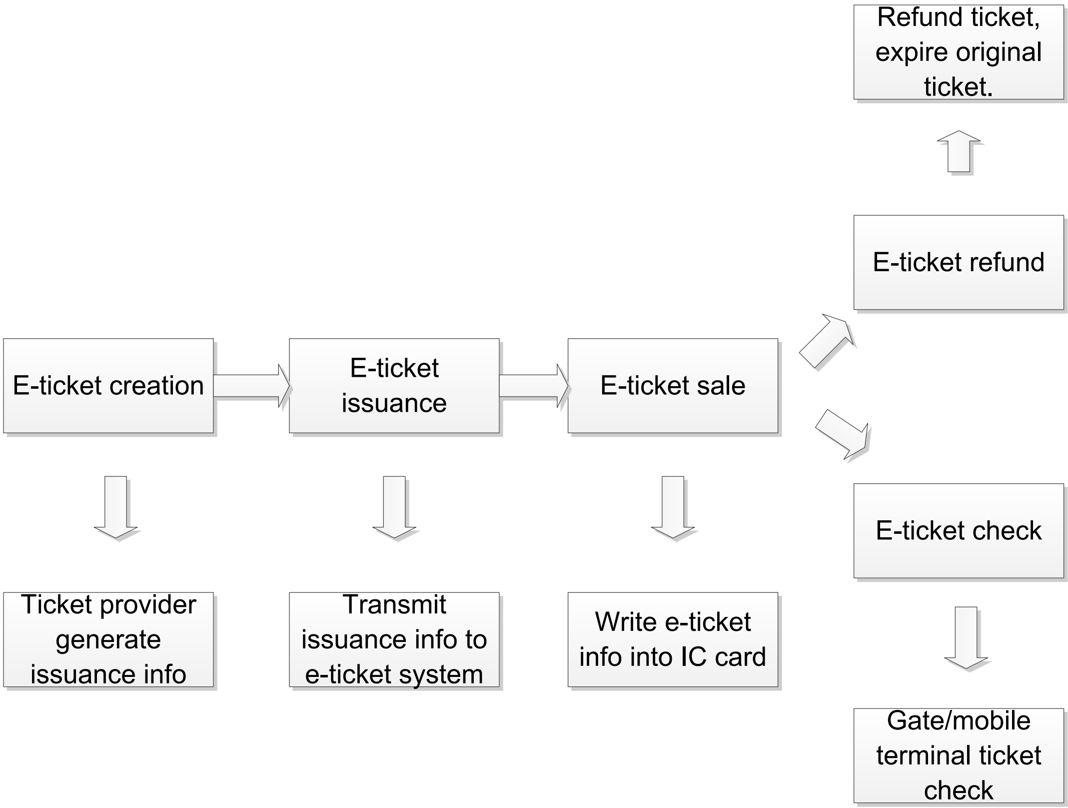

Figure A.5 E-ticket lifecycle chart

A.2.1.3 E-ticket Basic Functionality

1. E-ticket issuance

E-ticket issuance refers to e-ticket provider generating the issuance info, then batch transferring the digital info to the e-ticket system. Once the data reaches the e-ticket system, the issuance process is considered complete. In the subsequent ticket sale process, the e-ticket system must distribute the issuance info to the cards.

2. Pre-purchase browsing

Users can browse vendor provided e-ticket info through self-service terminals.

- Ticket provider websites are updated real-time with info of e-ticket available for sale, for users to search and browse

- Self-service terminal:

Ticket provider websites are updated real-time with info of e-ticket available for sale, for users to search and browse

3. E-ticket sale

E-tickets can support both on-site ticket sale and remote ticket sale.

- On-site ticket sale:

User can purchase tickets on site, after completing ticket payment through on site payment or other means, either service personnel uses ticket sale terminal to write e-ticket info into user card or use uses self-service terminal to write e-ticket into IC card.

- Remote ticket sale:

User can use internet to initiate ticket purchase request, and complete payment through UnionPay network. Use uses the network or internet terminal to write e-ticket info into card. All ticket data are stored on one IC card.

4. E-ticket check

E-tickets can utilize single use ticket checking, meaning that user holds the IC card containing the e-ticket, checking the ticket at the gate (or mobile ticket checking terminal). Gate verifies that the e-ticket info is valid and allows passage, with no need to check ticket as the e-ticket is removed.

- For one card one ticket users, user can check ticket directly;

- Gate (or mobile ticket checking terminal) gives corresponding indication of ticket check result (success or failure);

- For discount tickets, gate (or mobile ticket terminal) shall give special indication in the course of ticket checking.

5. Batch uploading of ticket checking info

Ticket check gates and mobile ticket check terminals will make daily batch uploads of ticket checking info to e-ticket system and ticket provider system.

6. E-ticket refund

Where permitted, user can process e-ticket refunds at designated service locations. Location customer service personnel use ticket sales terminal to verify that the e-ticket has not been used, issues refund, and uses ticket sales terminal to invalidate the original ticket.

7. E-ticket local administration

User can use the service website provided by the e-ticket provider and internet terminal to perform relevant administrative operations on e-tickets in cards. Including browsing purchased, expired, and used e-ticket info, as well as select or cancel "active" e-tickets.

A.2.2 E-ticket File Definition

In common e-ticket applications, there are two required pre-set e-ticket info data files, one is an open access e-ticket data file, and the other is a e-ticket file with key protected extension field. Main e-ticket data format requirements are as follows:

Table A.5 E-ticket data info file 1

File ID (SFI)			‘01H ’
File ID (FID)			EF01
File Type			Fixed Length Record File
File size			'194*10' decimal
File Access Control		Read = FREE	Write = FREE
Field	Date element	Format	Length
1	E-ticket merchant code	an	15
2	E-ticket provider code	ans	8
3	E-ticket serial no.	an	10

4	E-ticket name	ans	20
5	E-ticket info template type	B	1
6	E-ticket info	an	140

Table A.6 E-ticket data info file 2

File ID (SFI)			‘02H ’
File ID (FID)			EF02
File Type			Fixed Length Record File with Extension Field
File size			'194*10' decimal
File Access Control		Read = FREE	Write = maintenance key
Field	Date element	Format	Length
1	E-ticket merchant code	an	15
2	E-ticket provider code	ans	8
3	E-ticket serial no.	an	10
4	E-ticket name	ans	20
5	E-ticket info template type	B	1
6	E-ticket info	an	140

E-ticket information template type: Merchant chooses appropriate information template record e-ticket information based on specific business needs and e-ticket type.

01: Entrance ticket type

02: Movie ticket type

03: Merchant self-defined type Other: Reserved

01: Entrance type e-ticket info data file Entrance type e-ticket is used mainly as the electronic proof of a ticket holder's right to enter a restricted area. Relevant information, mainly including the usage format, type, corresponding ticket holder info, usage time, and corresponding rights of the e-ticket. File definition is as in the table below:

Table A.7 Entrance type e-ticket definition

Field	Date element	Length	Format	Code Description
1	Ticket type	3	an	'1XX'=registered name '2XX'=no registered name ’X1X’=single use ‘X2X’=multiple use ’XX1’=personal‘XX2’=group
2	Use name	20	ans	
3	Personal ID yype	1	B	
4	Personal ID number	20	an	
5	Designated time of day	6	n	hhmmsshhmmss, all 0 indicates no designated time of day.

6	Designated date range	8	n	YYYYMMDDYYYYMMDD, all 0 indicates no designated date range.
7	Corresponding counter serial number	1	B	
8	Group headcount	2	n	
9	Ticket check info	1	B	OA, unchecked, OB, checked
10	Ticket refund info	1	B	OA, not refunded, OB, refunded
11	Verification info	8	an	
12	Other	69	ans	

02. Movie type e-ticket info data file

Movie type e-ticket is an electronic proof of a ticket holder's right to enter a certain area and enjoy certain designated service or obtain some designated asset. The information involved, besides the e-ticket usage method, type, corresponding ticket holder info, usage time, and corresponding rights, also includes relevant information on service to be enjoyed or asset to be obtained. File definition is as in the table below:

Table A.8 Movie type e-ticket definition

Field	Date element	Length	Forma t	Code Description
1	Ticket type	3	an	'1XX'=registered name '2XX'=no registered name ’X1X’=single use ‘X2X’=multiple use ’XX1’=personal‘XX2’=group
2	Use name	20	ans	
3	Personal ID yype	1	B	
4	Personal ID number	20	an	
5	Designated time of day	6	n	hhmmsshhmmss, all 0 indicates no designated time of day.
6	Designated date range	8	n	YYYYMMDDYYYYMMDD all 0 indicates no designated date range.
7	Corresponding counter serial number	1	B	
8	Group headcount	2	n	
9	Ticket check info	1	B	OA, unchecked, OB, checked
10	Ticket refund info	1	B	OA, not refunded, OB, refunded

Field	Date element	Length	Forma t	Code Description
11	Venue or provider name	10	ans	Ex. movie ticket is theatre name or code
12	Ticket service type code	2	B	Ex. movie ticket is seating class code
13	Service type name	10	ans	Ex. movie ticket is seating class name
14	Asset code	4	B	Ex. movie ticket is film code
15	Name of asset designated to be included in ticket	16	ans	Ex. movie ticket is film title
16	Venue internal seating info	2	B	First byte indicates row no., last byte indicates serial no.
17	Designated service or product code	4	B	
18	Designated service or product description	16	ans	
19	Designated service or product price	4	B	
20	Designated service or product discount	1	n	
21	Verification code	8	an	

03: Issuer self-defined type e-ticket

Table A.9 Self-defined e-ticket definition

Field	Date element	Length	Format	Code Description
1	Ticket type length	1	n	
2	Ticket type	3	an	'1XX'=registered name '2XX'=no registered name 'X1X'=single use 'X2X'=multiple use 'XX1'=personal'XX2'=group
3	Corresponding counter serial number	1	B	
3	E-ticket body length	1	n	
4	E-ticket body (defined by issuer)	135	Ans	

If e-ticket needs to support multiple usage modes, then it must also support the counter file in the table below.

Table A.10 Counter file format

File ID (SFI)	'20' (decimal)
File ID (FID)	0020
File Type	Counter File

Counter number			20
File Access Control		Increment= increment key	Decrement= decrement key
Field	Date element		Length
1	Counter value 1		4
……			4
20	Counter value 20		4

The number of counters in the counter file is set to be 20. When each e-ticket is issued, as needed, an unused counter can be selected for use, and associate this data item through the counter serial number in the information file. E-ticket related proprietary keys mainly include 3 types - application control key, e-ticket info file application maintenance key, and the increment/decrement e-ticket counter key, with specific key info as follows:

Table A.11 Application control key:

Key Type			00
Key ID			00
Target access control		Read = Not allowed	Write = Application control key
Content	Application control key ACK value		Double length symmetric key

E-ticket increment key

Table A.12 Increment key format

Key Type	01

Key ID			0x56—0xff
Target access control		Read = Not allowed	Write = Application control key
Content	Increment key value		Double length symmetric key

E-ticket decrement key

Table A.13 Decrement key format

Key Type			01
Key ID			0x56—0xff
Target access control		Read = Not allowed	Write = Application control key
Content	Decrement key value		Double length symmetric key

A.2.3 E-ticket Personalization

Common data store applications that support e-tickets will co-exist in the same multi-application financial IC card with PBOC2.0 financial applications, and can be created ahead of time.

A.2.4 E-ticket usage flow example

A.2.4.1 Ticket sale flow

1. Remote ticket sale flow

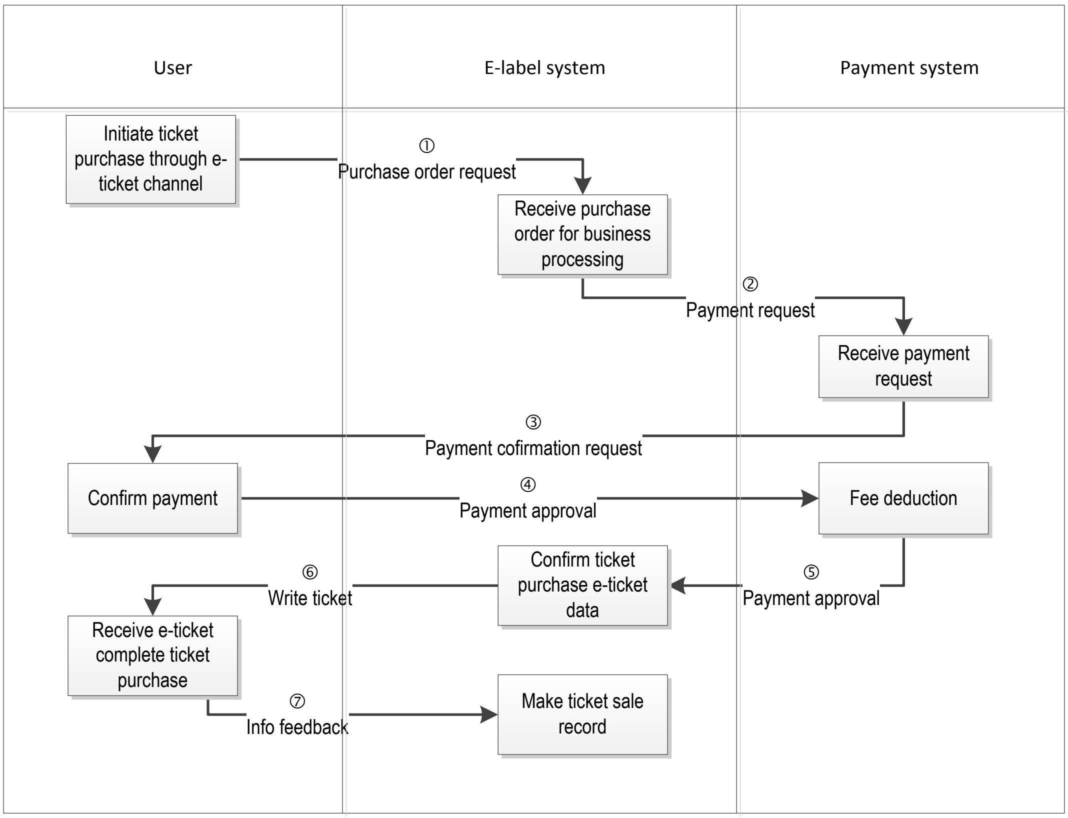

Figure A.6 Remote ticket purchase flow chart

(1) User initiates ticket purchase through internet;

(2) After e-ticket system receives user ticket purchase request, initiate payment request to the payment system based on the user selected mode of payment;

(3) After the payment system receives the e-ticket system payment request, initiate payment confirmation request to the user;

(4) User confirms payment;

(5) Payment system completes fee deduction, returns payment confirmation to e-ticket system;

(6) E-ticket system generates e-ticket data, sends to terminal, and under the protection of the corresponding maintenance key, writes the e-ticket info data file into the IC card. If it is a multi-use ticket, then use increment key to simultaneously update counter file;

(7) E-ticket system receives success confirmation data, records this ticket sale transaction.

2. On-site ticket sale flow

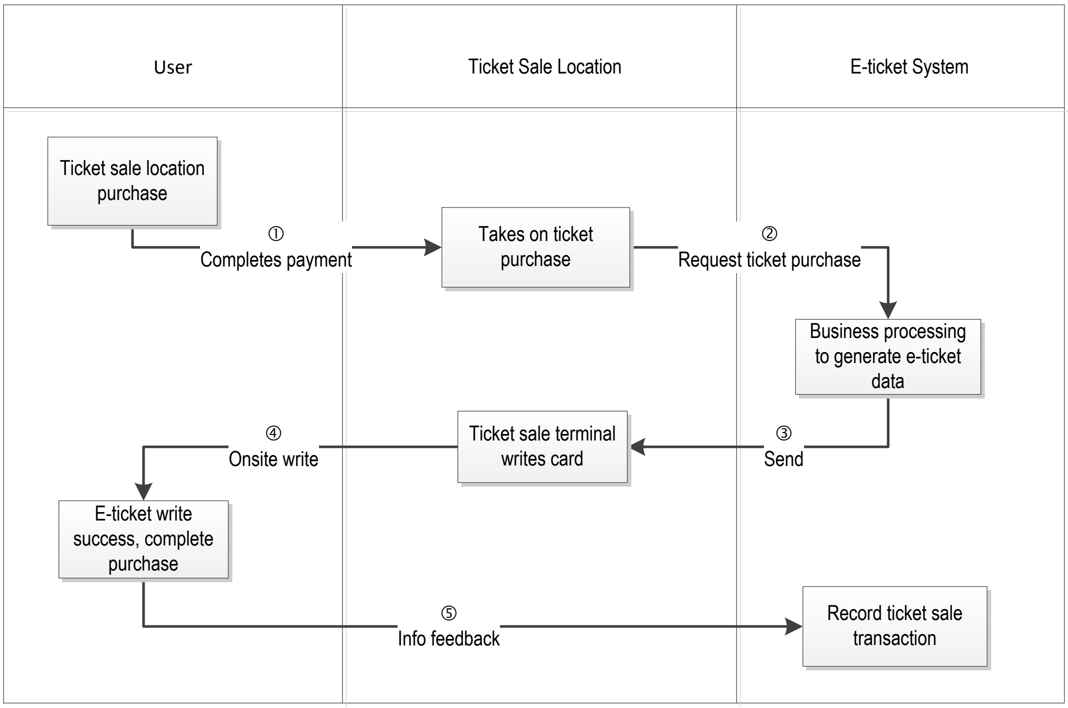

Figure A.7 Onsite ticket purchase flow chart

(1) User goes to ticket sale location, completes payment using cash, card, or other means of purchase;

(2) Service personnel initiates ticket purchase request through ticket sale terminal to the e-ticket system;

(3) E-ticket system generates e-ticket data, sends to ticket sale terminal;

(4) Under the protection of the corresponding maintenance key, ticket sales terminal writes the e-ticket info data file into the IC card. If it is a multi-use ticket, then use increment key to simultaneously update counter file;

(5) E-ticket system receives ticket write success confirmation data, records this transaction.

A.2.4.2 Replacement ticket sale flow

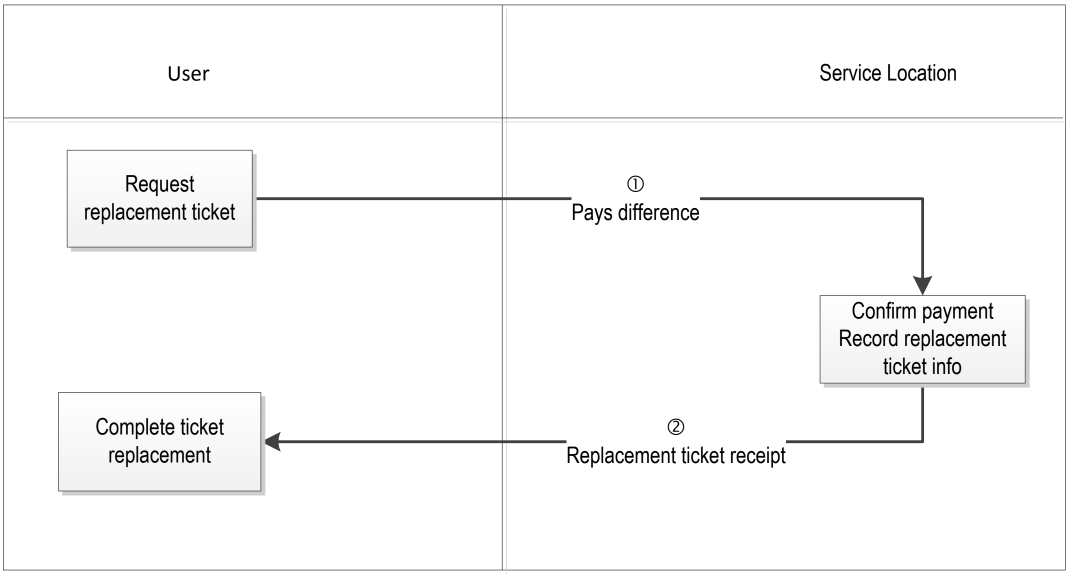

Figure A.8 Replacement ticket sale flow chart

(1) User requests replacement at service location, pays difference in ticket amount;

(2) Staff records replacement ticket info, gives user paper receipt and allows passage. Although this process does not involve specific operations on IC card or card applications, it is also a necessary processing flow for e-ticket usage.

A.2.4.3 Ticket checking flow

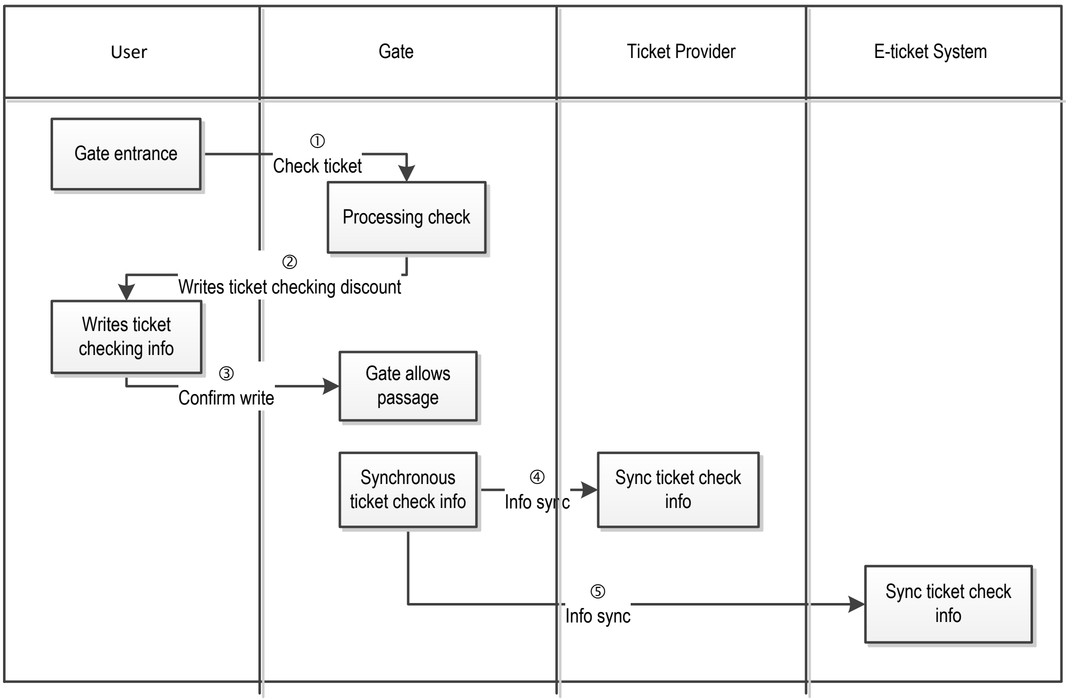

Figure A.9 Ticket checking flow chart

(1) User uses e-ticket checking at service area entrance;

(2) Ticket checking gate or mobile ticket checking terminal processes ticket checking, reads the data content from e-ticket card info file, checks for match and rights. For multi-use tickets, the terminal uses the counter serial number read from the info file to check whether the remaining uses in the counter meet usage requirements. Finally, use terminal secure storage designated maintenance key to update the ticket check info in the e-ticket IC card info data file. For multiple-use tickets, the terminal uses the counter serial number read from the info file to use the decrement key to deduct the usage times from the designated counter;

(3) Gate or mobile ticket checking terminal receives written confirmation for ticket checking info, allows passage;

(4) At end of day, the gate or mobile ticket checking terminal synchronizes ticket checking info to the ticket provider system;

(5) At end of day, the gate or mobile ticket checking terminal synchronizes ticket checking info to the e-ticket system;

A.3 Membership Card

A.3.1 Membership Card Functionality Intro

By putting member applications on chip, the common data store specification can be implemented with multi-member administration, providing cardholders with multiple functionality through a single card. Typical member card application types may include: aviation member card, fund member card, purchase member card, and travel member card:

A.3.2 Member card application file design

Member card application is one of the basic functionality for fast common application directories. Relevant support application elementary files can be pre-set in fast common application directories. There are two types of member card application data files, one is an open access member info file, and the other is a key protected member info file. In addition, point accumulation counter with key protection may be associated as needed. As shown below:

——Member info file

Member info file 1 indicates writing member info that has relatively low security grade requirements. Generally does not require security protection.

Table A.14 Member card info file 1

File ID (SFI)			‘01h’
File ID (FID)			EF01h
File Type			Fixed Length Record File
File size			'95*10' decimal
File Access Control		Read = FREE	Write = FREE
Field	Date element	Format	Length
1	Joint merchant code	an	15

2	Joint provider code	ans	8
3	Joint member card number	an	10
4	Member card name	ans	20
5	Member info template type	B	1
6	Member info (info template included data)		41

Member card info file 2 refers to fixed length record files with extended protected field.

Table A.15 Member card info file 2

File ID (SFI)			'02h'
File ID (FID)			EF02h
File Type			Fixed Length Record File with Extension Field
File size			'95*10' decimal
File Access Control		Read = FREE	Write = maintenance key
Field	Date element	Format	Length
1	Joint merchant code	an	15
2	Joint provider code	ans	8
3	Joint member card number	an	10

4	Member card name	ans	20
5	Member info template type	B	1
6	Member info (info template included data)		41

The member info template types mentioned in the table above are as follows:

01 Ordinary member info template

Table A.16 Ordinary member info

Field	Date element	Length	Format	Code Description
1	Member name	15	ans	
2	Member class	4	an	
3	Member discount rate	1	n	
4	Member card effective start date	4	B	YYYYMMDD, all 0 indicates no effective start date.
5	Member card expiration date	4	B	YYYYMMDD, all 0 indicates no expiration date.
6	Reserved bytes	13		

02 Member info template with point accumulation or mileage content.

Table A.17 Member info with point accumulation or mileage

Field	Date element	Length	Format	Code Description
1	Member name	15	ans	
2	Member class	4	an	
3	Member discount rate	1	n	
4	Member card effective start date	4	B	YYYYMMDD, all 0 indicates no effective start date.
5	Member card expiration date	4	B	YYYYMMDD, all 0 indicates no expiration date.
6	Member card verification key id (if no need to support then set to 0xFF)	1	B	
7	Point, mileage counter serial number	2	B	00 no associated counter
8	SFI of point/mileage counter file and PIN verification ID (if no need to support points then set to 0xFF)	1	B	Bit8=1, indicates decrement operation requires PIN verification; bit7=1, indicates increment operation requires PIN authentication; bit6: reserve for future use; bit 5~1: SFI
9	Reserved bytes	9		

03 Self-defined member info template

Table A.18 Self-defined member info template

Field	Date element	Length	Format	Code Description
1	Member name	15	ans	
2	Member class	4	an	
9	Reserved bytes	22		

Note: For member info file data items or reserved bytes, if there is no corresponding data to store, or data length is less than designated length, then 0XFF is used to right fill until the requirement length is reached. Member card verification key id corresponds to an internal authentication key. For joint parties that require authentication of member card authenticity, a key value can be set so that authenticity can be verified when member card is used.

——Point accumulation counter file

Some joint cards require implementation of point accumulation administrative function, for example purchase point accumulation, offline usage point (for gift exchange etc.), expired reset, and other operations, therefore the following counter files must also be supported, with each counter file corresponding to one joint party point accumulation application, and with the joint party managing the increment key and decrement key. This file corresponds to the first 5 records in member info file 02.

Point accumulation counter file 1

Table A.19 Point accumulation counter file 1

File ID (SFI)	11h
File ID (FID)	EF11h
File Type	Counter File
Counter number	1

File Access Control		Increment= increment key	Decrement= decrement key
Field	Date element		Length
1	Counter value 1		4

Point accumulation counter file 2

Table A.20 Point accumulation counter file 2

File ID (SFI)			12h
File ID (FID)			EF12h
File Type			Counter File
Counter number			1
File Access Control		Increment= increment key	Decrement= decrement key
Field	Date element		Length
1	Counter value 1		4

Point accumulation counter file 3

Table A.21 Point accumulation counter file 3

File ID (SFI)	13h
File ID (FID)	EF13h
File Type	Counter File

Counter number		1
File Access Control	Increment= increment key	Decrement= decrement key
Field	Date element	Length
1	Counter value 1	4

Point accumulation counter file 4

Table A.22 Point accumulation counter file 4

File ID (SFI)		14h
File ID (FID)		EF14h
File Type		Counter File
Counter number		1
File Access Control	Increment= increment key	Decrement= decrement key
Field	Date element	Length
1	Counter value 1	4

Point accumulation counter file 5

Table A.23 Point accumulation counter file 5

File ID (SFI)	15h
File ID (FID)	EF15h

File Type		Counter File
Counter number		1
File Access Control	Increment= increment key	Decrement= decrement key
Field	Date element	Length
1	Counter value 1	4

——Key management

Member card application related proprietary keys mainly include 3 types - application control key, member info file application maintenance key, and increment/decrement related keys that control the point counter. Based on the number of reserved member info files, corresponding number of keys can be redefined, with specific key info as follows:

Application control key

Table A.24 Application control key

Key Type		00
Key ID		00
Target access control	Read = Not allowed	Write = Application control key
Content	Application control key ACK value	Double length symmetric key

Application maintenance key

Table A.25 Application maintenance key

Key Type	01

Key ID		01-05
Target access control	Read = Not allowed	Write = Application control key
Content	Member card application maintenance key value	Double length symmetric key

Increment key

Table A.26 Increment key

Key Type		01
Key ID		0x56—0xff
Target access control	Read = Not allowed	Write = Application control key
Content	Increment key value	Double length symmetric key

Decrement key

Table A.27 Decrement key

Key Type		01
Key ID		0x56—0xff
Target access control	Read = Not allowed	Write = Application control key
Content	Decrement key value	Double length symmetric key

Internal authentication key

Table A.28 Internal authentication key

<table>
<tr><td colspan="3">Key Type</td><td>01</td></tr>
<tr><td colspan="3">Key ID</td><td>01-05</td></tr>
<tr><td colspan="2">Target access control</td><td>Read = Not allowed</td><td>Write = Application control key</td></tr>
<tr><td>Content</td><td colspan="2">Member card application internal authentication key value</td><td>Double length symmetric key</td></tr>
</table>

A.3.3 Member Card Application Personalization

Common data store applications that support member cards will co-exist in the same multi-application financial IC card with PBOC2.0 financial applications, and can be created ahead of time.

A.3.4 Member Card Application Usage Flow Example

Operating flow and terminal -- card command flow example:

1. Add member application

Adding member application means adding a new member record info in the member info file. The operating flow is as follows:

(1) Card holder inserts card;

(2) Verify cardholder offline PIN (or other means of authenticating rights);

(3) Write record into card according to pre-determined member info file format, with application maintenance key protection for the write operation.

Command Flow Explanation

a) Select member card application (select);

b) PIN verification, adding member info record requires cardholder agreement (verify pin);

c) Add record command, record can be added after passing cardholder verification. Based on security requirements, adding record requires MAC calculation (Append E-Record).

2. Use member application

Using member application means the merchant reading member application info (including member card no., effective date, counter serial number, and counter SFI), then based on purchase situation, increment or decrement member points. The operating flow is as follows:

(1) Card holder inserts card;

(2) Verify cardholder offline PIN (or other means of authenticating rights);

(3) Each merchant terminal searches its own org member card, reads card member info file, verify member info (card no., etc.), and apply corresponding discount to user purchase;

(4) Verify member card authenticity (optional);

(5) At the same time, update the corresponding point accumulation counter, so that with the next purchase, the user can self-select purchase points to in lieu of cash.

Command Flow Explanation

a) Select member application (select);

b) PIN verification, using member application requires cardholder agreement (verify pin);

c) Each merchant terminal searches its own org member card record (Search record), reads record command, reads all member application info (read record);

d) Internal verification (optional), verify whether member application is recognized by merchant, rejecting fake member card numbers (internal authenticate)

e) Purchase complete, based on member point management make corresponding increment or decrement point operations to counter file member points, if necessary update member application record, for example member effective date etc. (Increase Count, Decrease Count, Update E-Record).

3. Member application deletion

When the member application data store space is full, user can select to delete some members. The operating flow is as follows:

(1) Card holder inserts card;

(2) Verify cardholder offline PIN and authenticate control key, then obtain the right to delete records;

(3) Terminal read all user member info, and delete corresponding member application as designated by user.

Command Flow Explanation

a) Select member card application (select);

b) PIN verification, deleting member application requires cardholder agreement (verify pin);

c) Read record command, reads all member application info (read e-record);

d) Per user designated member application, delete the corresponding member records and clear the corresponding points accumulated (Delete E-Record, Decrease Count).

Appendix B (Normative Annex)
Bundling Rules for Record Files with Extension Field and Counter

B.1 Bundled Counter File Requirements

1. Counter SFI is 0x10, if ADF already contains files with SFI of 0x10, then card returns 6985;

2. The number of counters in the counter file is equal to the number of records in the record file with extension field. The record number in the record file with extension field corresponds to the counter serial number in the counter file;

3. Counter increment key must be obtained by using the counter file increment key for key derivation factoring;

4. Counter decrement key is the record maintenance key for the corresponding record;

5. In the counter file, after the extension field record is deleted, the corresponding counter is also reset. Counter state is set to be invalid, and invalid state counters cannot be used;

6. When record in record file with extension field successfully APPEND, the corresponding counter is simultaneously initialized. Initialized counters can be individually incremented or decremented.

B.2 Counter Increment Key Derivation

Counter increment key must be obtained by using the counter file increment key for key derivation factoring;

B.2.1 Key Derivation Factor Calculation

Take 8 bytes of all 0, using the corresponding record's record maintenance key for 3DES encryption. The obtained encryption result is the counter increment key's key derivation factor.

B.2.2 Increment Key Derivation Algorithm

The method to infer the left half of the counter increment key is:

——Use the derivation factor obtained in B.2.1 as input data;

——Use the counter file increment key as the encryption key;

——Use the counter file increment key to run 3DEA calculation on the input data. The method to infer the right half of the counter increment key is:

——Invert the derivation factor obtained in B.2.1 as input data;

——Use the counter file increment key as the encryption key;

——Use the counter file increment key to run 3DEA calculation on the input data.

Appendix C (Normative Annex)
Detailed explanation of financial IC card common application data store command response codes

This section mainly defines the reasons for error corresponding to some response codes.

C.1 Install Parameter

SW1	SW2	Reason for Error
6A	80	1. Common data store space to be created is greater than 0x7FFF
		2. DACK key authentication limit is 0; or greater than 15, and not equal to 0xFF
		3. DAAK key authentication limit is 0; or greater than 15, and not equal to 0xFF
		4. PIN key authentication limit is 0; or greater than 15, and not equal to 0xFF
		5. DDF name (common data store name) is greater than 16 bytes
6A	84	1. Insufficient space on card when creating common data store space
6D	00	1. Command does not exist

C.2 Create File Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error

SW1	SW2	Reason for Error
67	00	1. LC length error
69	82	1. Create file does not meet security state
		2. MAC calculation error
69	84	1. CLA=04, and no random number generated
		2. Insufficient space to create ADF file
		3. Insufficient space to create EF file
6A	80	1. Create EF file, SFI>30 or SFI=0
		2. 0 space to create ADF file
		3. File to be created already exist
		4. File AID to be created already exist
		5. 0 space to create binary file or is greater than 510 bytes
		6. Create fixed length record file, number of records or record length is 0
		7. Create circular record file, number of records or record length is 0
		8. Create variable length record file, record file size is 0
		9. Create counter file, number of counters is 0
		10. Create security file, file ID is not EF00

SW1	SW2	Reason for Error
		11. Create security file, number of keys is 0
		12. Create extended record file, number of records or record length is 0
		13. When creating ADF, defined application block/unblock key ID is incorrect
		14. When creating ADF, abnormal data format (no 4F, aid dfname length incorrect, ACK retry value error, ACK initial value error)
		15. Create record file with extension field, record content greater than 230
69	85	1. Create EF03 in common data store space
		2. Create FCI-SFI file defined in ADF but not binary file
6A	81	1. Created file type is not supported

C.3 Delete DF Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
69	82	1. Does not meet delete DF file rights
6A	86	1. P1=0, current directory is ADF file
		2. P1=1, current directory is ADF file

SW1	SW2	Reason for Error
67	00	1. LC error
6A	82	1. P1=0, DF file to be deleted does not exist

C.4 Read Count Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
67	00	1. LC length error
6A	86	1. P1, P2 error
6A	82	1. Counter file to be read does not exist
69	81	1. Command and file structure mismatch
69	84	1. Counter number to be searched for is equal to or less than 0
6A	83	1. Counter number to be searched for greater than the number of counters

C.5 Increase Count Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error

SW1	SW2	Reason for Error
6A	82	1. Counter file to be read does not exist
69	81	1. Command and file structure mismatch
69	84	1. No random number generated
69	88	1. MAC error
94	02	1. Counter overflow
94	03	1. Search for security file failure

C.6 Decrease Count Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
6A	82	1. Counter file to be read does not exist
69	81	1. Command and file structure mismatch
69	84	1. No random number generated
69	88	1. MAC error
94	02	1. Counter overflow
94	03	1. Search for security file failure

C.7 Get Data Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error

C.8 Write Key Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
69	84	1. No random number generated
69	85	1. Write DRPK and PIN into ADF
69	88	1. MAC error
94	03	1. Search for security file failure
6A	80	1. P1=0, add type 0 key
		2. P1=0, tried to add already existing key (type and ID are identical)
		3. P1=1, update key type is 0, and not DACK, DAAK, or ACK key

SW1	SW2	Reason for Error
		4. Key data field does not meet requirements (external authentication key Limit greater than 0x0f or less than 0x00, external authentication of key where second or third bytes are not 0x00)
		5. Pin value error
6A	84	1. Security file number of keys is maxed out
6A	83	1. P1=1, key to be updated not found

C.9 Get Challenge Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC length error

C.10 Internal Authenticate Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC length error
69	81	1. Command and file structure mismatch
		1. Search for security file failure

SW1	SW2	Reason for Error
94	03	2. Key not found
6A	82	1. Search for extended record file failure
6A	83	1. Extended record file not usable

C.11 External Authenticate Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC length error
69	84	1. No random number generated
69	83	1. Authentication key is blocked
69	88	1. Security message invalid
63	CX	1. Authentication failure, X indicates retries remaining
94	03	1. Search for security file failure
		2. Key not found

C.12 Read Binary Command

SW1	SW2	Reason for Error
6E	00	1. CLA error

SW1	SW2	Reason for Error
6A	82	1. Binary file to be read does not exist
6A	86	1. P1, P2 error
69	81	1. Command and file structure mismatch
69	82	1. Security status not met
6B	00	1. Offset error, P2>transparent file size
67	00	1. Data encryptiont processing, not a whole number multiple of 8
6C	XX	1. Post-encryption data !=Le, return 6CXX
		2. P2 plus Le is greater than file length (XX indicates remaining data length)
94	03	1. Search for security file failure
		2. Key not found

C.13 Update Binary Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	82	1. Binary file to be read does not exist
6A	86	1. P1, P2 error
6A	80	1. Cryptogram data decryption error

SW1	SW2	Reason for Error
69	81	1. Command and file structure mismatch
69	82	1. Security status not met
6B	00	1. Offset error, P2>transparent file size
67	00	1. Data length greater than update data length
69	84	1. No random number generated
69	88	1. MAC error
94	03	1. Search for security file failure
		2. Key not found

C.14 Append Record Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
		2. Added record length does not equal record length
6A	82	1. Designated variable length record file or circular record file not found
69	81	1. Command and file structure mismatch

SW1	SW2	Reason for Error
69	82	1. Security status not met
6A	84	1. Insufficient space in variable length record file
69	84	1. No random number generated
69	88	1. MAC error
94	03	1. Search for security file failure
		2. Key not found

C.15 Read Record Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
6A	82	1. File not found
69	81	1. Command and file structure mismatch
69	82	1. Security status not met
6A	83	1. Record not found
67	00	1. Returned data (plaintext or cryptogram) does not equal Le
6C	XX	1. Indicates permitted return data length
69	84	1. No random number generated

SW1	SW2	Reason for Error
69	88	1. MAC error
94	03	1. Search for security file failure
		2. Key not found

C.16 Update Record Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
		2. Length of record to be updated does not equal record length
6A	82	1. File not found
69	85	1. Update EF03 in DDF
6A	80	1. Cryptogram data decryption error
69	81	1. Command and file structure mismatch
69	82	1. Security status not met
6A	83	1. Binary file update without append
6A	84	1. Insufficient space in variable length record file
69	84	1. No random number generated

SW1	SW2	Reason for Error
69	88	1. MAC error
94	03	1. Search for security file failure
		2. Key not found

C.17 Verify PIN Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
69	83	1. Key is blocked
63	CX	1. X indicates number of retries remaining

C.18 Change PIN Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
		2. LC<5 or LC>13
6A	80	1. PIN format does not meet requirement

SW1	SW2	Reason for Error
69	83	1. Key is blocked
63	CX	1. X indicates number of retries remaining
94	03	1. Executing ChangePin command with missing PIN

C.19 Reload PIN Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
		2. LC<6 or LC>10
94	03	1. Search for security file failure
		2. Key not found
69	83	1. Key is blocked
69	84	1. No random number generated
69	88	1. MAC error
6A	80	1. PIN format does not meet requirement

C.20 Select File Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
		2. LC>16 or LC<5
6A	81	1. ADF application is blocked
6A	82	1. File not found

C.21 Get Response Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6D	00	1. In non-contact circumstance
6A	86	1. P1, P2 error
6F	00	1. No return data
6C	XX	1. Indicates permitted return data length

C.22 Append E-Record Command

SW1	SW2	Reason for Error
6E	00	1. CLA error

SW1	SW2	Reason for Error
6A	86	1. P1, P2 error
67	00	1. LC error
		2. Length of record to be added is less than the record length
6A	82	1. Designated record file not found
6A	80	1. Key data error
6A	84	1. File records all filled up
69	81	1. Command and file structure mismatch
69	84	1. No random number generated
69	88	1. MAC error
94	03	1. Search for security file failure
		2. Key not found

C.23 Update E-Record Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
		2. Length of record to be added is less than the record length

SW1	SW2	Reason for Error
6A	82	1. Designated record file not found
6A	83	1. Record not found
69	81	1. Command and file structure mismatch
69	84	1. No random number generated
69	88	1. MAC error
6A	80	1. Length of record to be updated does not equal record file record length
		2. Key data error

C.24 Delete E-Record Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
6A	82	1. Designated record file not found
6A	83	1. Record not found
69	81	1. Command and file structure mismatch
69	82	1. Security status not met

SW1	SW2	Reason for Error
69	84	1. No random number generated
69	88	1. MAC error
6A	80	1. Length of record to be updated does not equal record file record length

C.25 Search Record Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
6A	82	1. Designated record file not found
69	81	1. Command and file structure mismatch
69	82	1. Security status not met
6A	83	1. Record not found
69	88	1. MAC error
6A	80	1. Length of record to be updated does not equal record file record length

C.26 Application Block Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error
69	84	1. No random number generated
69	85	1. Current application is DDF
69	82	1. Security status not met
6A	81	1. Application is blocked
94	03	1. Key not found
69	83	1. Authentication key is blocked
69	88	1. Security message invalid
63	CX	1. Authentication failure, X indicates retries remaining

C.27 Application Unblock Command

SW1	SW2	Reason for Error
6E	00	1. CLA error
6A	86	1. P1, P2 error
67	00	1. LC error

SW1	SW2	Reason for Error
69	84	1. No random number generated
69	82	1. Security status not met
6A	81	1. Application is blocked
94	03	1. Key not found
69	83	1. Authentication key is blocked
69	88	1. Security message invalid
63	CX	1. Authentication failure, X indicates retries remaining